

①. The regressions eqn are $3x+2y=26$ and $bx+y=31$ find the correlation coefficient between X and Y .

Solution: given $3x+2y=26$, $bx+ay=31$

$$\begin{array}{r} 6x+y=31 \\ 6x+4y=52 \end{array}$$

$$-3y = -21$$

$$y = 7$$

$$6x+7=31$$

$$6x = 24$$

$$x = 4$$

$$\begin{array}{r} 52 \\ 31 \\ \hline 21 \end{array}$$

$$\begin{array}{r} 2 \\ 3 \times 7 \\ \hline 21 \end{array}$$

$$\textcircled{1} \Rightarrow 3x+2y=26$$

$$3x = 26 - 2y$$

$$x = \frac{26}{3} - \frac{2}{3}y$$

$$bry = -2/3$$

$$\textcircled{2} \Rightarrow bxy = 31$$

$$y = 31 - bx$$

$$byx = -b$$

$$y = \pm \sqrt{bxy/bx}$$

$$\therefore r = \pm \sqrt{b_{xy} \cdot b_{yx}}$$

$$= \pm \sqrt{\frac{-2}{3} \times -6}$$

$$= \pm \sqrt{4} = \pm 2$$

It is not possible because $= 1$ or ≤ 1

So,

$$\textcircled{1} \Rightarrow ay = 26 - 3x$$

$$y = \frac{26}{2} - \frac{3}{2}x$$

$$\boxed{b_{yx} = -3/2}$$

$$\textcircled{2} \Rightarrow bx + y = 31$$

$$bx = 31 - y$$

$$x = \frac{31 - y}{b}$$

$$\boxed{b_{xy} = -1/6}$$

$$r = \pm \sqrt{b_{xy} \cdot b_{yx}}$$

$$= \pm \sqrt{-1/2 \cdot -3/2}$$

$$= \pm \sqrt{1/4} = 1/2 \approx 0.5$$

Rank Correlation

The

$$r = 1 - \frac{\sum_{i=1}^n d_i^2}{n(n^2-1)}$$

Find the following d

Rank in X

Rank in Y

X	Y
1	4
2	3
3	1
4	2
5	6
6	5
7	7

Karl's pear

Wkt, 18 an

$r =$

$$r = 1 \text{ or } \leq 1$$

$$\begin{aligned} bx + y &= 31 \\ bx &= 31 - y \\ x &= \frac{31 - y}{b} \\ &= -\frac{1}{b} \end{aligned}$$

$$= 20.5$$

Rank Correlation Coefficient (or) Karl's Pearson Formula

The Rank correlation Coefficient

$$r = 1 - \frac{\sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Find the Rank correlation from the following data.

Rank in X	1	2	3	4	5	6	7
Rank in Y	3	1	2	6	5	7	

X	Y	$d_i = X - Y$	d_i^2
1	4	-3	9
2	3	-1	1
3	1	2	4
4	2	2	4
5	6	-1	1
6	5	1	1
7	7	0	0

$$\sum d_i^2 = 20$$

Karl's Pearson formula:

Wkt, Rank correlation,

$$r = 1 - \frac{\sum_{i=1}^n d_i^2}{n(n^2 - 1)} = 1 - \frac{20}{336}$$

$$\frac{48}{336} = \frac{1}{7}$$

$$= \frac{1-180}{336}$$

$$= \frac{336-180}{336}$$

$$\frac{336}{180}$$

$$= \frac{216}{336}$$

$$\frac{168}{336}$$

$$84$$

$$42$$

$$14$$

$$= \frac{9}{14}$$

$$= 0.642$$

$$\begin{array}{r} 0.642 \\ 14 \overline{) 9.0} \\ \underline{84} \\ 60 \\ \underline{56} \\ 40 \end{array}$$

$$1.086$$

$$0.357$$

$$0.693$$

Correlation coefficient for difference in
Variables X and Y,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

$$2\sigma_x\sigma_y$$

①. for the
most logical
madras corre
at bombay
price 68 at

Avg price
S.D of price

estimated of
S.D of the
price at end
solution

let $x =$

let $y =$

given that,

$$\bar{x} = 6$$

$$\sigma_x =$$

$$\sigma_{x-y}$$

$$\begin{array}{r} 336 \\ 120 \\ \hline 216 \\ 8 \overline{) 84279} \\ \underline{672} \\ 1707 \\ \underline{1360} \\ 3479 \\ \underline{3360} \\ 119 \end{array}$$

$$\begin{array}{r} 0.642 \\ 14 \overline{) 90} \\ \underline{84} \\ 60 \\ \underline{56} \\ 40 \\ \underline{38} \\ 20 \\ \underline{18} \\ 20 \\ \underline{18} \\ 20 \\ \underline{18} \\ 20 \\ \underline{18} \\ 20 \end{array}$$

$$\begin{array}{r} 0.693 \\ \hline 0.693 \end{array}$$

$$\overline{x-y}^2$$

①. for the following data find the most likely price at madras corresponding to the price 70 at bombay corresponding to the price 68 at madras?

	madras	Bombay
Avg price	65	67
S.D of price	0.5	0.3

standard deviation of the difference between the price at madras and bombay at 3.1

Solution

let x denote the price at madras
let y denote the price at Bombay

given that,

$$\bar{x} = 65, \bar{y} = 67$$

$$\sigma_x = 0.5, \sigma_y = 0.3$$

$$\sigma_{x-y} = 3.1$$

$$V = \sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2$$

$$2\sigma_x\sigma_y$$

$$= 0.25 + 12.25 - \sigma_{x-y}^2$$

$$0.30$$

$$= 0.25 + 12.25 - 9.61$$

$$2.89$$

$$= \frac{2.89}{3.5} = 0.8257$$

$$\begin{array}{r} 0.11 \\ 12.25 \\ 9.61 \\ \hline 2.64 \\ 0.25 \\ \hline 2.89 \end{array}$$

The line of regression y on x is

$$y - 67 = 0.8257 \cdot \frac{3.5}{0.5} (x - 65)$$

$$\begin{array}{r} 0.5 \\ 0.5 \\ \hline 2.5 \end{array}$$

$$\begin{array}{r} 0.0 \\ 0.0 \\ \hline 0.0 \end{array}$$

$$\begin{array}{r} 3.5 \\ 3.5 \\ \hline 1.75 \\ 1.05 \\ \hline 1.225 \end{array}$$

$$\begin{array}{r} 0.5 \\ 0.5 \\ \hline 1.5 \end{array}$$

$$\begin{array}{r} 0.0 \\ 0.15 \\ \hline 2.1 \end{array}$$

$$0.30$$

$$\begin{array}{r} 3.1 \\ 3.1 \\ \hline 3.1 \end{array}$$

$$\begin{array}{r} 9.3 \\ 9.3 \\ \hline 9.61 \end{array}$$

$$y - 67 = 0$$

$$y - 67 =$$

$$y = 5$$

Put $x =$

$$y =$$

$$y =$$

$$\boxed{y =}$$

$\therefore$ The
68 at ma
84.339

H.W. The

X on Y i

$$X - 6$$

$$X - 6$$

Put $Y = 70$

$$X - 6$$

$$X -$$

$$\begin{array}{r} 0.5 \\ 0.5 \\ \hline 2.5 \\ 0.0 \\ \hline 0.05 \end{array}$$

$$\begin{array}{r} 3.5 \\ 3.5 \\ \hline 1.75 \\ 1.05 \\ \hline 1.225 \end{array}$$

$$\begin{array}{r} 0.5 \\ 0.5 \\ \hline 1.5 \\ 0.0 \\ \hline 0.15 \\ 0.1 \\ \hline 0.30 \end{array}$$

$$\begin{array}{r} 31 \\ 31 \\ \hline 93 \\ 93 \\ \hline 961 \end{array}$$

$$x - 65$$

$$y - 67 = 0.8257 \cdot 7 (x - 65)$$

$$y - 67 = 0.8257 \cdot 7 (x - 65)$$

$$y = 5.7799 (x - 65) + 67$$

$$\text{Put } x = 68,$$

$$y = 5.7799 (3) + 67$$

$$y = 17.3397 + 67$$

$$\boxed{y = 84.3397}$$

$$\begin{array}{r} 5.7799 \\ 1.73397 \\ \hline 67 \\ \hline 174.464 \end{array}$$

$\therefore$ The corresponding to the price 68 at madras likely price at bombay is 84.3397 //

H.W The line of regression of $(x - \bar{x})$ or $\frac{\sigma_x}{\sigma_y} (y - \bar{y})$
X on Y is,

$$x - 65 = 0.8257 \cdot \frac{3.5}{0.5} (y - 67)$$

$$x - 65 = 0.8257 \cdot 7 (y - 67)$$

$$\text{Put } y = 70$$

$$x - 65 = 0.8257 \cdot 7 (3)$$

$$x - 65 = 17.3397 \Rightarrow x = 65 + 17.3397$$

$$x = 82.3397$$

idhu
dhoopu
Anu: 67
5597

Proof:

Transformation of a random variable

Let X, Y be a continuous two dimensional random variable with JPDF $f_{xy}(x, y)$ transforms X and Y to new random variable $U = h(x, y)$,
 $V = g(x, y)$

①. Let
dimens
 $f(x, y)$

find x
given

Dimensional
and $0 \leq$
~~let~~ $= 4n$

let $u =$

$u^2 =$

$x^2 =$

$x =$

$f(x)$

$$f_{uv}(u,v) = |J| f_{xy}(x,y)$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

① Let X, Y be a continuous two dimensional random variable with JPDF

$$f(x,y) = 4xy e^{-(x^2+y^2)}, \quad x \geq 0, y \geq 0$$

find the pdf $\sqrt{x^2+y^2}$

given,

X, Y is a continuous two

dimensional random variable in $0 \leq x < \infty$

and $0 \leq y < \infty$ also given that $f(x,y)$

$$f(x,y) = 4xy e^{-(x^2+y^2)}$$

$$\text{let } U = \sqrt{x^2+y^2} \quad V = y$$

$$U^2 = x^2 + y^2$$

$$x^2 = U^2 - V^2$$

$$x = \sqrt{U^2 - V^2}$$

$$f(x,y) = 4xy e^{-(x^2+y^2)}$$

2 Variable

0

ith JPDF

to

$$J = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$\frac{dx}{du} = \frac{1}{\sqrt{u^2 - v^2}} \quad \frac{dx}{dv} = \frac{u}{\sqrt{u^2 - v^2}}$$

$$\frac{dx}{dv} = \frac{-v}{\sqrt{u^2 - v^2}}$$

$$\frac{dy}{du} = 0, \quad \frac{dy}{dv} = 1$$

$$J = \begin{vmatrix} \frac{1}{\sqrt{u^2 - v^2}} & \frac{u}{\sqrt{u^2 - v^2}} \\ 0 & 1 \end{vmatrix}$$

$$= \frac{1}{\sqrt{u^2 - v^2}} - 0$$

$$= \frac{1}{\sqrt{u^2 - v^2}}$$

$f(u, v)$

Change

$x \geq$

$\sqrt{u^2 - v^2}$

$u^2 - v^2$

$u^2 \geq$

$u \geq$

Compare

$\therefore f(u, v)$

The pdf

$f(u)$

$$f_{uv}(u,v) = |J| f_{xy}(x,y)$$

$$= \frac{u}{\sqrt{u^2 - v^2}} \cdot 4 \sqrt{u^2 - v^2} v e^{-u^2}$$

$$= 4uv e^{-u^2}$$

Change the range for u and v .

$$x \geq 0, y \geq 0.$$

$$\sqrt{u^2 - v^2} \geq 0 \quad v \geq 0$$

$$u^2 - v^2 \geq 0 \quad v \geq 0$$

$$u^2 \geq v^2$$

$$u \geq v$$

Comparing $0 \leq v \leq u < \infty$.

$$\therefore f_{uv}(u,v) = 4uv e^{-u^2}, \quad 0 \leq v \leq u < \infty$$

The pdf of u is given by,

$$f_u(u) = \int_{v=0}^u f_{uv}(u,v) dv$$

$$= \int_0^u 4uv e^{-u^2} dv$$

$$= 4ue^{-u^2} \int_0^u v dv$$

$$= 4ue^{-u^2} \left[\frac{v^2}{2} \right]_0^u$$

$$= 2ue^{-u^2} u^2$$

$$= 2u^3 e^{-u^2} //$$

②. If X and Y are the independent random variable with pdf e^{-x} , $x \geq 0$, e^{-y} , $y \geq 0$ respectively find the

$$\text{pdf } U = \frac{X}{X+Y}, V = X+Y.$$

are U & V are independent

So if X and Y are so,

$$f(x)f(y) = f(x,y)$$

$$f(x,y) = e^{-x-y}$$

$$\text{Let } U = \frac{X}{X+Y}$$

$$U = \frac{X}{X+Y}$$

$$U = \frac{X}{V}$$

$$\boxed{X = UV}$$

$$J = \begin{vmatrix} \frac{dx}{du} \\ \frac{dy}{du} \end{vmatrix}$$

$$\frac{dx}{du} = v$$

$$\frac{dy}{du} = 0$$

$$J = \begin{vmatrix} v \\ 0 \end{vmatrix}$$

So if X and Y are independent variables

$$V \quad f(x) f(y) = f(x, y)$$

$$f(x, y) = e^{-x} e^{-y} = e^{-(x+y)}; x \geq 0, y \geq 0.$$

$$\text{let } U = \frac{x}{x+y}; \quad V = x+y$$

$$U = \frac{x}{x+y} \quad Y = v-x$$

$$U = \frac{x}{v} \quad ; \quad Y = v-x$$

$$\boxed{X = UV}$$

$$\boxed{Y = V - X}$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\frac{\partial x}{\partial u} = v \quad \left| \quad \frac{\partial x}{\partial v} = u \right.$$

$$\frac{\partial y}{\partial u} = 0 \quad \left| \quad \frac{\partial y}{\partial v} = 1 - u \right.$$

$$J = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v$$

x independent

$$f(x) = e^{-x}, x \geq 0$$

find the

$+ y$.

t

$$f_{uv} = |J| f_{xy}(x, y) = v e^{-(x+y)} = v e^{-(uv+u-v)} \\ = v e^{-(uv+u-v)} \\ = v e^{-u} //$$

Range for $u \geq v$,

$$x \geq 0$$

$$u \geq 0$$

$$v \geq 0$$

$$y \geq 0$$

$$v - uv \geq 0$$

$$v \geq uv$$

$$1 \geq u$$

PDF of U is given by, $0 \leq v \leq u \leq 1$

$$f_U(u) = \int_0^u f_{uv}(u, v) dv = \int_0^u v e^{-u} dv \\ = \left[v \left(\frac{e^{-u}}{-1} \right) - 1 \left(\frac{e^{-u}}{-1} \right) \right]_0^u \\ = \left[-v e^{-u} - e^{-u} \right]_0^u = 0 + 0 + e^{-u} = e^{-u}$$

PDF of V is given by

$$f_V(v) = \int_v^1 f_{uv}(u, v) du$$

$$= \int_v^1 v e^{-u} du$$

$$= v e^{-u} (1 - v)$$

To find U

to prove,

$$f(u) f(v)$$

$$f(u)$$

$\therefore U$ and

H.W.

①. If the

2-D R.V. (X, Y)

$0 \leq X, Y \leq 1$

Soln:

Given

R.V. defined

Also, given

$f_{xy}(x, y)$

we have

$$= ye^{-(xy)} = ye^{-0 \cdot xy}$$

$$y \geq 0$$

$$-0 \cdot y \geq 0$$

$$y \geq 0 \cdot y$$

$$\geq 0$$

$$\leq 1$$

$$\leq 1$$

$$= \int_0^{\infty} ye^{-v} dv$$

$$\int_0^{\infty}$$

$$+ 0 \cdot e^0 = 1$$

$$= \int_0^1 ye^{-v} dv = ye^{-v} \Big|_0^1 = ye^{-v} [v]_0^1$$

$$= ye^{-v} (1-0) = ye^{-v}$$

To find U and V are independent i.e.,
to prove,

$$f(u) f(v) = f(u, v)$$

$$f(u) f(v) = 1 \times ye^{-v} = ye^{-v} = f(u, v)$$

$\therefore U$ and V are independent.

H.W.

①. If the JPDF of the continuous
2-D R.V. (X, Y) given by $f_{XY}(x, y) = xy$,
 $0 \leq x, y \leq 1$ the PDF of $U = XY$.

Soln:

Given (X, Y) is a continuous 2D
R.V. defined in $0 \leq x \leq 1$ and $0 \leq y \leq 1$
Also, given that

$$f_{XY}(x, y) = xy, \quad 0 \leq x, y \leq 1$$

we have to find $u = xy$ — ①



let $v=y \Rightarrow y=v$

① $\Rightarrow u=xv \Rightarrow x=\frac{u}{v}, y=v$

$$\frac{\partial x}{\partial u} = \frac{1}{v} ; \quad \frac{\partial x}{\partial v} = \frac{-u}{v^2}$$

$$\frac{\partial y}{\partial u} = 0 ; \quad \frac{\partial y}{\partial v} = 1$$

$$J = \begin{vmatrix} \frac{1}{v} & \frac{-u}{v^2} \\ 0 & 1 \end{vmatrix}$$

The JPDF of (u,v) $f_{uv}(u,v) = |J| f_{xy}(x,y)$

$$= \left| \frac{1}{v} \right| (x,y) = \frac{1}{v} \left(\frac{u}{v} + v \right)$$

$$f_{uv}(u,v) = \left(\frac{u}{v^2} + 1 \right)$$

To find the range for $u \neq v$:

$$\text{we have } 0 \leq x \leq 1 \Rightarrow 0 \leq \frac{u}{v} \leq 1$$

$$\Rightarrow 0 \leq u \leq v$$

$$\text{Also, } 0 < y < 1 \Rightarrow 0 < v < 1$$

Combining $0 \leq$

we get $0 \leq$

$$\therefore f_{uv}(u,v)$$

PDF of u,v

$$f_u(u) = \int_v$$

$$= \int \frac{uv^{-2+1}}{-2+1}$$

$$= -u + 1$$

$$= -2u$$

$$(x,y)$$

2 Rand

Combining $0 \leq u \leq v \leq 1$ and $0 \leq v \leq 1$

we get $0 \leq u \leq v \leq 1$

$$\therefore f_{uv}(u, v) = \frac{u}{v^2} + 1, \quad 0 \leq u \leq v \leq 1$$

PDF of u is given by,

$$f_u(u) = \int_u^1 f_{uv}(u, v) dv = \int_u^1 \left(\frac{u}{v^2} + 1 \right) dv$$

$$= \left[\frac{uv^{-2+1}}{-2+1} + v \right]_u^1 = \left[\frac{-u}{v} + v \right]_u^1$$

$$= -u + 1 + \frac{u}{u} - u$$

$$= -2u + 2 \Rightarrow 2(1-u); \quad 0 \leq u \leq 1$$

$\therefore (X, Y)$ are continuous.

2) Random process:

$u \neq v:$

$$\frac{u}{v} \leq 1$$

Unit-3

Random Process

Def:

A random process is a collection of random variable $\{X(s, t)\}$ that of functions of a real variable. namely time "t". where $s \in S, t \in T$.

State space:

The set of all possible value of any member of the random process is called state space.

Stationary Process (or) strict sense stationary process

If a random process is stationary

Formula:
 $E[X(t)] = \mu$

Formula:
standard

then

- 1)
- 2)

3)

4)

5)

If a random process is stationary to all order then the random process is said to be strict sense

stationary.

Formula:
 $E[X(t)] = \text{constant}$ $V[X(t)] = \text{constant}$

Formula:

standard integral formula:

if n is a integer, $n \neq 0$

then, 1) $\int_0^{2\pi} \cos n\theta d\theta = 0$

2) $\int_{-\pi}^{\pi} \cos n\theta d\theta = 0$

3) $\int_0^{2\pi} \cos(\omega t + n\theta) d\theta = 0$

4) $\int_{-\pi}^{\pi} \cos(\omega t + n\theta) d\theta = 0$

5) $\int_0^{2\pi} \sin n\theta d\theta = 0$

$$6) \int_{-\pi}^{\pi} \sin n\theta d\theta = 0.$$

$$7) \int_0^{2\pi} \sin(\omega t + n\theta) d\theta = 0.$$

$$8) \int_{-\pi}^{\pi} \sin(\omega t + n\theta) d\theta = 0.$$

$$9) \int_0^{2\pi} d\theta = 2\pi.$$

$$10) \int_{-\pi}^{\pi} d\theta = 2\pi.$$

$$11) \int_{-\pi}^{\pi} \cos(\omega t + \theta) d\theta = -2 \sin \omega t.$$

Discrete Random Variable: ~~not~~
 Q. If the random process with probability 1/2

Q. If the random process $X(t)$ that the value -1 with probability $1/3$ and takes the value 1 with probability $2/3$. find whether $X(t)$ is a stationary process or not.

Soln:

given

To prove
 i.e., to

(i) E

(ii) V

(i) E

(i) $X(t)$

$E[X]$

Soln:

given that

$X(t) = n$	-1	1
$P(n)$	$1/3$	$2/3$

To prove $X(t)$ is a ss process
ie, to prove

(i) $E[X(t)] = \text{constant}$

(ii) $V[X(t)] = \text{constant}$

(i) $E[X(t)] = \sum n p(n)$

$$= -1(P(-1)) + 1P(1)$$

$$= -1 \times \frac{1}{3} + 1 \times \frac{2}{3} = -\frac{1}{3} + \frac{2}{3}$$

$$= \frac{1}{3} = \text{constant}$$

(ii) $\text{Var}[X(t)] = E[X^2(t)] - (E[X(t)])^2$

$$E[X^2(t)] = \sum n^2 p(n)$$

$$= (-1)^2 P(-1) + (1)^2 P(1)$$

$$= \frac{1}{3} + \frac{2}{3} = \frac{3}{3} = 1$$

$$\begin{aligned}\text{Var}[X(t)] &= 1 - \left(\frac{1}{3}\right)^2 \\ &= 1 - \frac{1}{9} = \frac{8}{9} //\end{aligned}$$

2) The process $\{X(t)\}$ whose probability distribution under certain condition is given by $P\{X(t)=n\} = \frac{(at)^{n-1}}{(1+at)^n}$,

$n=1, 2, 3, \dots$

$$= \frac{at}{1+at}, \quad n=0$$

so that it is not stationary (evolutionary).

The probability distribution fn of

$X(t)=n$	0	1	2	3	...
$P(n)$	$\frac{at}{1+at}$	$\frac{1}{(1+at)^2}$	$\frac{at^2}{(1+at)^3}$	$\frac{(at)^2}{(1+at)^4}$	$\dots$

To prove $X(t)$ is a stationary (or) SSS process

(i) $E[X(t)] = \text{const}$
 (ii) $V[X(t)] = \text{const}$

i) $E[X(t)] = \sum_{n=0}^{\infty} n p(n)$

$= 1 \times P(1) + 2P(2) + 3P(3) + \dots$

$= \frac{1}{(1+at)^2} + \frac{2at}{(1+at)^3} + \frac{3(at)^2}{(1+at)^4} + \dots$

$1 + 3at + 3a^2t^2 = (1+at)^2$

$= \frac{1}{(1+at)^2} \left[1 + \frac{2at}{1+at} + \frac{3(at)^2}{(1+at)^2} + \dots \right]$

$E[X(t)] = \frac{1}{(1+at)^2} \left[(1 + \frac{at}{1+at})^{-2} \right]$

$= \frac{1}{(1+at)^2} \left[\frac{1+at-at}{(1+at)} \right]^{-2}$

$= \frac{1}{(1+at)^2} \left[\frac{1}{1+at} \right]^{-2}$

$= \frac{1}{(1+at)^2} \times (1+at)^2 = \text{constant}$

$$(ii) V[X(t)] = E[X^2(t)] - (E[X(t)])^2$$

$$E[X^2(t)] = \sum_{n=0}^{\infty} n^2 p(n)$$

$$= \sum_{n=0}^{\infty} (n^2 + n - n) p(n)$$

$$= \sum_{n=0}^{\infty} (n^2 + n) p(n) - \sum_{n=0}^{\infty} n p(n)$$

$$= \sum_{n=0}^{\infty} n(n+1) p(n) - 1$$

$$= 1 \times 2 p(1) + 6 \times p(2) + 12 p(3) + \dots$$

$$= 1 \times \frac{2}{(1+at)^2} + 6 \times \frac{at}{(1+at)^3} + 12 \times \frac{(at)^2}{(1+at)^4} + \dots$$

$$- 1$$

$$= \frac{2}{(1+at)^2} \left[1 + \frac{3at}{(1+at)} + \frac{6(at)^2}{(1+at)^2} + \dots \right]$$

$$= 1$$

$$= \frac{2}{(1+at)^2}$$

$$= \frac{2}{(1+at)^2}$$

$$= \frac{2}{(1+at)^2}$$

$$= 0$$

$$= 0$$

$$V[X(t)]$$

1)]] ^2

$$= \frac{2}{(1+at)^2} \left[1 - \frac{at}{1+at} \right]^3 - 1$$

$$= \frac{2}{(1+at)^2} \left[\frac{1+at-at}{1+at} \right]^3 - 1$$

$$= \frac{2}{(1+at)^2} \times (1+at)^3$$

$$= 2(1+at) - 1$$

$$= 2 + 2at - 1 = 1 + 2at$$

$$V[X(t)] = 1 + 2at$$

$$= 2at \neq \text{constant}$$

$\therefore$ If is not stationary

Hence proved it is Evolutionary

WSS process : (wide sense stationary)

A random process $x(t)$ is said to be wide sense stationary. If it satisfies the following condition

(i) $E[x(t)] = \text{constant}$

(ii) $R(t_1, t_2) = \text{a function of } t_1 - t_2$

1) Show that the random process $x(t) = A \cos(\omega t + \theta)$

is wide sense stationary if A and ω are constant and θ is uniformly distributed random variable in $(0, 2\pi)$.

Given that,

$$x(t) = A \cos(\omega t + \theta)$$

where θ is uniformly distributed in $(0, 2\pi)$

$$f(\theta) = \begin{cases} \frac{1}{2\pi} & ; 0 \leq \theta \leq 2\pi \\ 0 & ; \text{otherwise} \end{cases}$$

$$= \frac{1}{2\pi}$$

to prove

i) $E[x]$

(ii) $R(t)$

(i) $E[x]$

$$= \int_0^{2\pi}$$

$$= \frac{A}{2\pi}$$

$$= \frac{A}{2\pi}$$

(ii) R

$$= E[x^2]$$

$$= A^2$$

$$= A^2$$

stationary)

is process
satisfies
in

of

$t = t_2$

$\cos(\omega t + \theta)$ is

distributed

$\int$

$\sum$

trans

$$= \frac{1}{2\pi} \quad ; \quad 0 \leq \theta \leq 2\pi$$

to prove $x(t)$ is a WSS process

i) $E[x(t)] = \text{constant}$.

(ii) $R(t_1, t_2) = \text{a function of } t_1 - t_2$.

(i) $E[x(t)] = \int_0^{2\pi} x(t) f(\theta) d\theta$

$$= \int_0^{2\pi} A \cos(\omega t + \theta) \frac{1}{2\pi} d\theta$$

$$= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega t + \theta) d\theta$$

$$\int_0^{2\pi} \cos(\omega t + \theta) d\theta = 0$$

$$= \frac{A}{2\pi} \times 0 = 0 = \text{constant}$$

(ii) $R[t_1, t_2] = E[x(t_1) x(t_2)]$

$$= E[A \cos(\omega t_1 + \theta) A \cos(\omega t_2 + \theta)]$$

$$= A^2 E[\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)]$$

$$= A^2 E\left[\frac{1}{2} \cos(\omega t_1 + \theta + \omega t_2 + \theta) + \frac{1}{2} \cos(\omega t_1 + \theta - \omega t_2 - \theta)\right]$$

$$= \frac{A^2 E}{2} \left[\cos(\omega t_1 + \omega t_2 + 2\theta) + \cos(\omega(t_1 - t_2)) \right]$$

$$= \frac{A^2 E}{2} \left[\cos(\omega t_1 + \omega t_2 + 2\theta) \right] + \frac{A^2 E}{2} \left[\cos \omega(t_1 - t_2) \right]$$

~~$$= \frac{A^2 E}{2} \left[\cos \right]$$~~

$$= \frac{A^2}{2} \int_0^{2\pi} \cos(\omega t_1 + \omega t_2 + 2\theta) \times \frac{1}{2\pi} d\theta$$

$$+ \frac{A^2}{2} \cos(\omega(t_1 - t_2))$$

$$= \frac{A^2}{2\pi} \times \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega t_1 + \omega t_2 + 2\theta) d\theta$$

$$+ \frac{A^2}{2} \cos(\omega(t_1 - t_2))$$

$$= \frac{A^2}{2\pi} \times 0 + \frac{A^2}{2} \cos \omega(t_1 - t_2)$$

$$= \frac{A^2}{2} \cos \omega(t_1 - t_2) = \text{function of } t$$

$X(t)$ is a wss process

H.W

If ~~X(t)~~ X

is a r.v.
are indeps
examine

(i) find f_X

$$X(t) =$$

$$E[X(t)]$$

$\therefore M_X$

(ii) find

$R_X(t)$

sub

$$R_X(t)$$

$$= 1$$

HW

If $X(t) = A \cos \lambda t + B \sin \lambda t, t \geq 0$
is a random process where A and B
are independent $N(0, \infty)$ random variables
examine the wss property of $X(t)$.

(i) Find the Mean,

$$X(t) = A \cos \lambda t + B \sin \lambda t$$

$$\begin{aligned} E[X(t)] &= E[A] \cos \lambda t + E[B] \sin \lambda t \\ &= 0 \times \cos \lambda t + 0 \times \sin \lambda t \\ &= 0 \end{aligned}$$

$\therefore$ Mean is Constant.

(ii) Find the Auto correlation Function,

$$R_X(t_1, t_2) = E[X(t_1) X(t_2)]$$

Sub $X(t_1)$ and $X(t_2)$

$$\begin{aligned} R_X(t_1, t_2) &= E[(A \cos \lambda t_1 + B \sin \lambda t_1) \\ &\quad (A \cos \lambda t_2 + B \sin \lambda t_2)] \\ &= E[A^2] \cos \lambda t_1 \cos \lambda t_2 + E[AB] \cos \lambda t_1 \sin \lambda t_2 \\ &\quad + E[AB] \sin \lambda t_1 \cos \lambda t_2 + E[B^2] \sin \lambda t_1 \sin \lambda t_2 \end{aligned}$$

given,

$$E[A^2] = E[B^2] = \sigma^2, E[AB] = 0.$$

Hence,

$$R_X(t_1, t_2) = \sigma^2 \cos \lambda t_1 \cos \lambda t_2 + \sigma^2 \sin \lambda t_1 \sin \lambda t_2$$

take σ^2 common,

$$R_X(t_1, t_2) = \sigma^2 [\cos \lambda t_1 \cos \lambda t_2 + \sin \lambda t_1 \sin \lambda t_2]$$

Use formula

$$\cos A \cos B + \sin A \sin B = \cos(A - B)$$

We get

$$R_X(t_1, t_2) = \sigma^2 \cos \lambda(t_1 - t_2)$$

let

$$\tau = t_1 - t_2$$

$$R_X(\tau) = \sigma^2 \cos(\lambda \tau)$$

So the $X(t)$ is wide sense

stationary process.

2) $X(t) = V \cos$
are independent
processes, the
probability

show that
is it SSF

Soln:

give

$$\frac{U}{P(U)}$$

$$\frac{V}{P(V)}$$

E

E[

2) $X(t) = U \cos t + V \sin t$ where U and V are independent ^{random} variables, each of which assumes the values -2 and 1 with the probability $1/3$ and $2/3$ respectively. Show that $X(t)$ is a WSS process? Is it SS process?

Soln:

given that $X(t) = U \cos t + V \sin t$

U	-2	1
$P(U)$	$1/3$	$2/3$

V	-2	1
$P(V)$	$1/3$	$2/3$

$$E[U(t)] = \sum U P(U)$$

$$= -2 \times \frac{1}{3} + 1 \times \frac{2}{3}$$

$$E[U^2(t)] = \sum U^2 P(U) \quad \text{--- } 0 \text{ is constant}$$

$$= (-2)^2 \times \frac{1}{3} + 1^2 \times \frac{2}{3}$$

$$= \frac{4}{3} + \frac{2}{3} = 2$$

$$E(v) = \sum v P(v) = \frac{-2}{3} + \frac{2}{3} = 0$$

$$E[v^2(t)] = \sum v^2 P(v) = 2$$

given that u and v are independent random variable $\therefore E[uv] = E[u]E[v]$

$$= 0 \times 0 = 0$$

$$\therefore E[uv] = 0$$

To prove $E[X(t)]$ is a wss process that is to prove i) $E[X(t)] = \text{const}$

$$\text{ii) } R(t_1, t_2) =$$

a function of $t_1 - t_2$

$$E[X(t)] = E[\cos t + v \sin t]$$

$$= E[\cos t] + E[v \sin t]$$

$$= E[v] E[\cos t] + E[v]$$

$$= 0 + 0 = 0 \Rightarrow \text{constant } E[\sin t]$$

$$(i) R(t_1, t_2) = E[X(t_1) X(t_2)]$$

$$= E[(U \cos t_1 + V \sin t_1) (U \cos t_2 + V \sin t_2)]$$

$$= E[U^2 \cos t_1 \cos t_2 + V^2 \sin t_1 \sin t_2 + UV \cos t_1 \sin t_2 + UV \sin t_1 \cos t_2]$$

$$= E[U^2 \cos t_1 \cos t_2] + E[UV \cos t_1 \sin t_2] + E[UV \sin t_1 \cos t_2] + E[V^2 \sin t_1 \sin t_2]$$

$$= E[U^2] \cdot E[\cos t_1 \cos t_2] + E[UV] E[\cos t_1 \sin t_2] + E[UV] \cdot E[\sin t_1 \cos t_2] + E[V^2] E[\sin t_1 \sin t_2]$$

$$= 2 E[\cos t_1 \cos t_2] + 2 E[\sin t_1 \sin t_2]$$

$$= 2 \cos(t_1 - t_2) = \text{a function of } t_1 - t_2$$

$\therefore X(t)$ is a WSS process

SSS process: M.W.

$$V[X(t)] = E[X^2(t)] - (E[X(t)])^2$$

$$E[X^2(t)] = E[U \cos t + V \sin t]^2$$

$$= E[U^2 \cos^2 t + V^2 \sin^2 t + 2UV \cos t \sin t]$$

$$= E[U^2] \cos^2 t + E[V^2] \sin^2 t + 2 E[UV] \cos t \sin t$$

$$= 2 \cos^2 t + 2 \sin^2 t + 0$$

$$= 2 (\cos^2 t + \sin^2 t)$$

$$= \underline{\underline{2}}$$

$$V[X(t)] = E[X^2(t)] - (E[X(t)])^2$$

$$= 2 - 0^2$$

$$= 2$$

Poisson

event (random Poisson

$$P[X(t)]$$

(i) Sup

a bank

a mean

Probability

interval

(ii) M

H.W. (iii) for

let X

i) P(x)

Poisson Process:

the occurrence of a certain event $(0, +\infty)$ than the discrete random process $\{X(t)\}$ is called Poisson process.

$$P[X(t) = n] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

(i) Suppose that the customers arrive at a bank according to the Poisson process with a mean rate of 3 p/m. find the probability that during the time interval of 2 minutes i) exactly 4 customers arrive

(ii) More than 4 customer arrive.

(iii) fewer than 4 customer in 2 minutes.
let $X(t)$ denote no. of customer arrive at a bank in the time interval of t .
 $t = 2$, $\lambda = 3$.

$$\begin{aligned} \text{i) } P(X=4) &= \frac{e^{-3 \times 2} (3 \times 2)^4}{4!} = \frac{e^{-6} \times 1296}{24} \\ &= e^{-6} \times 54 = 0.1338. \end{aligned}$$

(i) P[More than 4 customers]

$$= P[X(t) \geq 4]$$

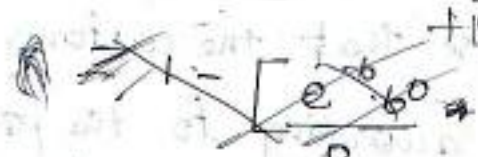
$$= 1 - P[X(t) \leq 4]$$

$$= 1 - P[X(t) = 0]$$

$$+ P[X(t) = 1] +$$

$$P[X(t) = 2] + P[X(t) = 3]$$

$$+ P[X(t) = 4]$$



$$= 1 - e^{-6} - e^{-6} \cdot 6 - e^{-6} \cdot \frac{6^2}{2!} - e^{-6} \cdot \frac{6^3}{3!} - e^{-6} \cdot \frac{6^4}{4!}$$

$$= 1 - 0.0148 - 0.0892 - 0.5354$$

$$= 1 - 0.0148$$

$$= 0.9852$$

$$= 0.9852$$

$$= 1 - 0.0148 - 0.0892 - 0.5354$$

$$= 0.3704$$

$$= 0.3704$$

(ii) P[fewer than

$$= P[X$$

$$= P[X$$

$$= e^{-1}$$

$$= e$$

$$= 6$$

$$= 6$$

MARKOV PRO

(given) p
set to be

$$P[X(t) \leq$$

MA RKO

process

7

$$x(t) \leq 4$$

$$x(t) \leq 4$$

$$x(t) = 0$$

$$x(t) = 1$$

$$x(t) = 2$$

$$x(t) = 3$$

$$x(t) = 4$$

$$x(t) = 5$$

$$x(t) = 6$$

$$x(t) = 7$$

$$x(t) = 8$$

$$x(t) = 9$$

$$x(t) = 10$$

$$x(t) = 11$$

$$x(t) = 12$$

(iii) P [fewer than 4 customers]

$$= P[X(t) \leq 4]$$

$$= P[X(t) = 0] + P[X(t) = 1] +$$

$$P[X(t) = 2] + P[X(t) = 3]$$

$$= e^{-6} \left[1 + 6 + \frac{36}{2} + \frac{216}{6} \right]$$

$$= e^{-6} [1 + 6 + 18 + 36]$$

$$= 61e^{-6}$$

$$= 0.1512$$

MARKOV PROCESS:

A random process $X(t)$ is set to be markov process, if

$$P[X(t) \leq x / X(t_1) = x_1, X(t_2) = x_2, \dots, X(t_{n-1}) = x_{n-1}]$$

$$= P[X(t) \leq x / X(t_n) = x_n]$$

MARKOV CHAIN:

A discrete parameter markov process is called Markov chain

$$P\{X_n = a_n / X_{n-1} = a_{n-1}, X_{n-2} = a_{n-2}, \dots, X_0 = a_0\}$$

$$\Rightarrow P\{X_n = a_n / X_{n-1} = a_{n-1}\} \text{ for all } n.$$

then the process $\{X_n, n = 0, 1, 2, \dots\}$ are called Markov chain. Here $a_1, a_2, \dots$ are called states of Markov chain.

Transition Probability Matrix: (TPM)

$$P = \{P_{ij}\} \text{ is called (one-step)}$$

transition probability.

$$P_{ij} = P\{X_{n+1} = j / X_n = i\}, 1 \leq i \leq m$$

$$\text{then } P = P[ij] = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{m1} & \dots & \dots & P_{mn} \end{bmatrix}$$

Steady state

$$* \pi P = \pi \text{ or}$$

$$* \pi = (\pi_1, \pi_2, \dots)$$

①. The transition matrix of a Markov chain

1, 2, 3 and

$$P[ij] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

find (i)

(ii)

Soln:

Given

2.

Steady state distribution: (Property)

* $\pi P = \pi$ and $\pi_1 + \pi_2 + \dots + \pi_k = 1$.

* $\pi = (\pi_1, \pi_2, \pi_3, \dots, \pi_k)$.

①. The transition probability matrix (TPM) of a Markov chain having 3 states.

1, 2, 3 and Initial Distribution is

$P(0) = (0.7, 0.2, 0.1)$ is

$P = \begin{pmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{pmatrix}$ 16 marks

find (i) $P[X_3=2, X_2=3, X_1=3, X_0=2]$.

(ii) $P[X_2=3]$.

Soln:

Given that, State space = $\{1, 2, 3\}$

$P = \begin{pmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{pmatrix}$

Initial Distribution,

$P(0) = (0.7, 0.2, 0.1)$

$P(X_0=1) = 0.7$ $P(X_0=2) = 0.2$

$P(X_0=3) = 0.1$

$$i) P[X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2]$$

$$= P[X_3 = 2 / X_2 = 3] \times P[X_2 = 3 / X_1 = 3] \times$$

$$P[X_1 = 3 / X_0 = 2] \times$$

$$= 0.0048 //$$

$$ii) P[X_2 = 3] = \sum_{i=1}^3 P[X_2 = 3 / X_0 = i] P[X_0 = i]$$

$$= P[X_2 = 3 / X_0 = 1] P[X_0 = 1]$$

$$+ P[X_2 = 3 / X_0 = 2] P[X_0 = 2] +$$

$$P[X_2 = 3 / X_0 = 3] P[X_0 = 3]$$

$$= P_{13}^{2-0} 0.7 + P_{23}^{2-0} 0.2 + P_{33}^{2-0} 0.1$$

$$= P_{13}^2 0.7 + P_{23}^2 0.2 + P_{33}^2 0.1$$

$$P^2 = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \times \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$$

$$= 0.01 + 0.36 + 0.7$$

$$= \begin{bmatrix} 0.01 + 0.3 + 0.12 & 0.05 + 0.1 + 0.16 & 0.04 + 0.1 + 0.12 \\ 0.06 + 0.12 + 0.06 & 0.3 + 0.04 + 0.08 & 0.24 + 0.04 + 0.06 \\ 0.03 + 0.24 + 0.09 & 0.15 + 0.08 + 0.12 & 0.24 + 0.08 + 0.09 \end{bmatrix}$$

$$= \begin{bmatrix} 0.43 & 0.31 & 0.26 \\ 0.24 & 0.42 & 0.34 \\ 0.36 & 0.35 & 0.29 \end{bmatrix}$$

$$P[X_2=3] = P_{13}^2 \cdot 0.1 + P_{23}^2 \cdot 0.2 + P_{33}^2 \cdot 0.1$$

$$= (0.26)(0.1) + (0.34)(0.2) + (0.29)(0.1)$$

$$= 0.182 + 0.068 + 0.029$$

$$= 0.279$$

$$\begin{bmatrix} 0.43 \\ 0.24 \\ 0.36 \end{bmatrix}$$

Auto Correlation:

$$R_{XX}(\tau) = E\{X(t)X(t+\tau)\}$$

where $X(t)$ is a random variable or process.

Property Remark:

$R_{XX}(\tau)$ is an even fn of τ i.e., $R_{XX}(\tau) = R_{XX}(-\tau)$.

Wkt,

$$R_{XX}(\tau) = E\{X(t)X(t+\tau)\}$$

$$R_{XX}(-\tau) = E\{X(t)X(t-\tau)\}$$

$$\text{Put } t-\tau = p$$

$$t = p + \tau$$

$$R_{XX}(-\tau) = E\{X(p+\tau)X(p)\}$$

$$= E\{X(p)X(p+\tau)\}$$

$$= R_{XX}(\tau)$$

property:

The maximum value of $R_{XX}(t)$ is attained at the point $t=0$, i.e.

$$|R_{XX}(t)| \leq R_{XX}(0);$$

prove that $R(t)$ is maximum at $t=0$

$$E[X(t_1) \pm X(t_2)]^2 \geq 0.$$

$$E[X^2(t_1)] + E[X^2(t_2)] \pm 2E[X(t_1)X(t_2)]$$

Since,

$$R_{XX}(0) = E[X^2(t)]$$

$$R_{XX}(0) + R_{XX}(0) + 2E[X(t_1)X(t_2)] \geq 0$$

$$2R_{XX}(0) + 2E[X(t_1)X(t_2)] \geq 0.$$

$$2R_{XX}(0) \pm 2R_{XX}(t_1, t_2) \geq 0$$

$$2R_{XX}(0) \geq 2R_{XX}(t_1, t_2)$$

$$\Rightarrow R_{XX}(0) \geq |R_{XX}(t)|$$

①. find the Mean & Variance of a stationary process whose Autocorrelation

fn is given by (i) $R(\tau) = 2 + 4e^{-2|\tau|}$.

$$(ii) R(\tau) = 2.5\tau^2 + 36$$

Soln:

$$R(\tau) = 2 + 4e^{-2|\tau|}$$

$$\mu_{x^2} = \lim_{\tau \rightarrow \infty} R(\tau)$$

$$= \lim_{\tau \rightarrow \infty} 2 + 4e^{-2|\tau|}$$

$$= 2 + 4e^{-\infty}$$

$$= 2 + 4 \times 0 = 2$$

$$\mu_x = \pm \sqrt{2} = \pm 1.414$$

$$\text{Mean} = E[X(t)] = \pm \sqrt{2}$$

$$E[X^2(t)] = \lim_{\tau \rightarrow 0} R(\tau)$$

$$= \lim_{\tau \rightarrow 0} 2 + 4e^{-2|\tau|}$$

$$= 6 //$$

$$\text{Var}[X(t)]$$

$$(ii) R(\tau)$$

$$\mu_{x^2}$$

$$= \lim_{\tau \rightarrow \infty} R(\tau)$$

$$E[X^2(t)]$$

$$\text{Var}[X(t)]$$

②.

The with E

find

Soln

$$\text{Var}[X(t)] = E[X^2(t)] - E[X(t)]^2$$

$$= 6 - (2)^2 = 4$$

$$(ii) R(z) = 25z^2 + 36$$

$$\mu_{x^2} = \lim_{z \rightarrow \infty} R(z)$$

$$= \lim_{z \rightarrow \infty} \frac{25z^2 + 36}{625z^2 + 4} = \lim_{z \rightarrow \infty} \frac{25z^2 + 36}{625z^2 + 4}$$

$$= \lim_{z \rightarrow \infty} \left(\frac{25 + \frac{36}{z^2}}{625 + \frac{4}{z^2}} \right) = \pm 2$$

$$E[X^2(t)] = \lim_{z \rightarrow \infty} R(z) = \lim_{z \rightarrow \infty} \frac{25z^2 + 36}{625z^2 + 4} = 9$$

$$\text{Var}[X(t)] = E[X^2(t)] - (E[X(t)])^2$$

$$= 9 - (2)^2 = 5$$

②. The Random process $X(t)$ is stationary with $E[X(t)] = 1$, $R(\tau) = 1 + e^{-2|\tau|}$
find the mean & variance of $S(t) = \int_0^t X(t) dt$

Soln:

Given,

$$E[X(t)] = 1$$

$$R(\tau) = 1 + e^{-2|\tau|}$$

$$S(t) = \int_0^t X(t) dt$$

To find,

$$\text{Mean of } S(t) = E[S(t)]$$

$$\text{Variance of } S(t) = E[S^2(t)] - (E[S(t)])^2$$

Mean of $S(t)$,

$$E[S(t)] = E\left[\int_0^1 X(t) dt\right]$$

$$= \int_0^1 E[X(t)] dt$$

$$= [t]_0^1 = 1$$

Variance of $S(t)$,

$$E[S^2(t)] = \int_{-1}^1 \int_{-1}^1 (1-|\tau|)R(\tau) d\tau$$

$$= \int_{-1}^1 \int_{-1}^1 (1-|\tau|)(1+e^{-2|\tau|}) d\tau$$

$$= 2 \int_0^1 (1-\tau)(1+e^{-2\tau}) d\tau$$

$$u = 1 - \tau$$

$$u' = -1$$

$$u'' = 0$$

$$dv = 1 + e^{-2\tau}$$

$$v = \tau + \frac{-e^{-2\tau}}{-2}$$

$$v_1 = \frac{\tau^2}{2} + \frac{e^{-2\tau}}{4}$$

$$E[S^2(t)]$$

$$= 2$$

$$= 2$$

$$= 2$$

$$= 2$$

$$\text{Var}[S]$$

$$= \frac{3}{2}$$

$$=$$

$$E[s^2(t)] = 2 \left[(1-\tau) \left(\tau - \frac{e^{-2\tau}}{2} \right) \right.$$

$$\left. - (-1) \left(\frac{\tau^2}{2} + \frac{e^{-2\tau}}{4} \right) \right]$$

$$= 2 \left[(1-1) \left(1 - \frac{e^{-2}}{2} \right) + \left(\frac{1}{2} + \frac{e^{-2}}{4} \right) \right.$$

$$\left. - (1-0) \left(0 - \frac{e^0}{2} \right) - \left(\frac{0}{2} + \frac{1}{4} \right) \right]$$

$$= 2 \left(\frac{1}{2} + \frac{e}{4} + \frac{1}{2} - \frac{1}{4} \right)$$

$$= 2 \left(\frac{3}{4} + \frac{e^{-2}}{4} \right) = \frac{3+e^{-2}}{2}$$

$$\text{Var}[s(t)] = E[s^2(t)] - (E[s(t)])^2$$

$$= \frac{3+e^{-2}}{2} - 1 = \frac{3+e^{-2}-2}{2}$$

$$= \frac{1+e^{-2}}{2} //$$

Cross Correlation: ~~(*)~~ ~~(*)~~

1) Let $x(t)$ and $y(t)$ be the 2 random process, then the cross correlation is defined as

$$R_{xy}(t, t+\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x y f(x, y; t, t+\tau) dx dy$$

Property:

if $x(t)$ & $y(t)$ are two random process $R_{xx}(\tau)$ & $R_{yy}(\tau)$

their respective Autocorrelation

fn then $|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0) R_{yy}(0)}$

①. Consider Two Random

process $x(t) = 3 \cos(\omega t + \theta_1)$ &

$y(t) = 2 \cos(\omega t + \theta_2)$

where θ is the random variable

Uniformly
that $|R_x$

Soln:

Given
in $(0, 2\pi)$

f

f

$x(t) =$

$x(t_1)$

$x(t_2)$

To find

$$= E[$$

$$= 9E[u$$

$$= \frac{9}{2} E$$

2
mass

Uniformly Distributed in $(0, 2\pi)$ prove
that $|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0) R_{yy}(0)}$

Soln:

$y = t$
 $+ \tau$
study.

Given θ is uniformly distributed
in $(0, 2\pi)$ $f(x) = \frac{1}{b-a}$

$$f(\theta) = \frac{1}{b-a} = \frac{1}{2\pi}$$

$$f(\theta) = \frac{1}{2\pi}, \quad 0 < \theta < 2\pi$$

$$X(t) = 3 \cos(\omega t + \theta)$$

$$X(t_1) = 3 \cos(\omega t_1 + \theta)$$

$$X(t_2) = 3 \cos(\omega t_2 + \theta)$$

$$\begin{aligned} \cos A \cos B &= \frac{1}{2} [\cos(A+B) \\ &+ \cos(A-B)] \end{aligned}$$

To find $R_{xx}(0)$:

$$R(t_1, t_2) = E[X(t_1) X(t_2)]$$

$$= E[3 \cos(\omega t_1 + \theta) * 3 \cos(\omega t_2 + \theta)]$$

$$= 9 E[\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)]$$

$$= \frac{9}{2} E[\cos(\omega t_1 + \theta + \omega t_2 + \theta) + \cos(\omega t_1 + \theta - \omega t_2 - \theta)]$$

$$= \frac{q}{2} E \left[\cos(\omega(t_1+t_2)+2\theta) + \cos(\omega(t_1-t_2)) \right]$$

$$R_{XX}(0,0) = \frac{q}{2} E [\cos(\omega(0)+2\theta) + \cos(0)]$$

$$= \frac{q}{2} E [\cos 2\theta + 1] + \frac{q}{2} E [1]$$

$$= \frac{q}{2} \int_0^{2\pi} \cos 2\theta f(\theta) d\theta + \frac{q}{2}$$

$$= q/2$$

likewise,

$$Y(t) = 2 \cos(\omega t + \theta - \pi/2)$$

$$Y(t_1) = 2 \cos(\omega t_1 + \theta - \pi/2)$$

$$Y(t_2) = 2 \cos(\omega t_2 + \theta - \pi/2)$$

$$R(t_1, t_2) = E[Y(t_1) \times Y(t_2)]$$

$$= E[2 \cos(\omega t_1 + \theta - \pi/2) \times 2 \cos(\omega t_2 + \theta - \pi/2)]$$

$$= q E [\cos$$

$$= \frac{4}{2} E [1$$

$$= 2 E [$$

$$= 2 E$$

$$= 2 E$$

$$R_{YY}(0$$

$$= 2 E$$

$$= 2 E$$

$$= 2$$

$$=$$

2) +

2)]

2) + cos 0]

2)]

$$= 4E [\cos(\omega t_1 + \theta - \pi/2) \cos(\omega t_2 + \theta)]$$

$$= \frac{4}{2} E [\cos(\omega t_1 + \theta - \pi/2 + \omega t_2 + \theta - \pi/2) + \cos(\omega t_1 + \theta - \pi/2 - \omega t_2 - \theta + \pi/2)]$$

$$= 2E [\cos(\omega(t_1 + t_2) + 2\theta - \pi) + \cos(0)]$$

$$= 2E [\cos(\omega(t_1 + t_2) + 2\theta - \pi/2) + 1]$$

$$= 2E [\cos(\omega(t_1 + t_2) + 2\theta - \pi/2)]$$

$$R_{YY}(0, 0) = 2E [\cos(\omega(0) + 2\theta - \pi/2)]$$

$$= 2E [1 + \cos(2\theta - \pi/2)]$$

$$= 2E [2 - \cos(\pi/2 + 2\theta)]$$

$$= 2E [2 - \int_0^{2\pi} \cos 2\theta f(\theta) d\theta]$$

$$= 2 - 0 = 2 //$$

③

$$|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)R_{yy}(0)}$$

$$R_{xy}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

$$= E[3\cos(\omega t_1 + \theta) \sin(\omega t_2 + \theta)]$$

$$= E[6\cos(\omega t_1 + \theta) \sin(\omega t_2 + \theta)]$$

$$= 6E[\sin(\omega(t_1 + t_2) + 2\theta) + \sin(\omega(t_1 - t_2) + 0)]$$

$$= 3E[\sin(\omega(t_1 + t_2) + 2\theta)] + 3E[\sin(\omega(t_1 - t_2))]$$

$$= 3 \int_0^{2\pi} \sin(\omega(t_1 + t_2) + 2\theta) \cdot \frac{1}{2\pi} d\theta + 3E[\sin(\omega(t_1 - t_2))]$$

$$= \frac{3}{2\pi} \times 0 + 3\sin(\omega\tau)$$

$$= 3\sin(\omega\tau)$$

$$= 3\sin \omega \tau //$$

$\therefore X(t)$ and $Y(t)$ are jointly

$$WSS \quad |R_{xy}(\tau)| = |3\sin(\omega\tau)| \leq 3$$

To prove:

$$\sqrt{R_{xx}(0)R_{yy}(0)} \geq |R_{xy}(\tau)|$$

$$\sqrt{R_{xx}(0)R_{yy}(0)}$$

Analy

A

Statistical

to check

more

different

2

3

1) Comp

Th

transfo

11.0
frame

$$\sqrt{R_{xy}(co) R_{yx}(co)} = \sqrt{\frac{9}{8} \times 2}$$

$$= 3$$

$$= |R_{xy}(co)| \quad //$$

5 Unit

Analysis of Variants: (ANOVA)

Analysis of variants is a statistical technique that is used to check if the mean of 2 or more groups are significantly different.

1 way - column or row

2 way - CCR

3 way $\rightarrow$ 1) + findout

1) Completely Randomized Design

one way classification
There are three main

transfer of certain counter a set of sample values is tested and found to be allocated among 4 grps

A, B, C, D and their branches, etc

Brands	Groups			
	A	B	C	D
I	10	4	8	15
II	5	8	13	6
IV	8	19	11	13

is there any significant difference in brand? assume at 5% level of significant.

Step 1:

Null Hypothesis:

H_0 : There is no significant difference between brands (rows)

Step 2:

Alternate Hypothesis

H_1 : There is ~~no~~ significant difference between brands (rows)

Step 2:

A.

H_1 :

A
B
C
D

Step 3: n

Step 4:

Step 5:

$$\begin{array}{r} 110 \\ 110 \\ \hline 000 \\ 100 \\ \hline 110 \\ 100 \\ \hline 220 \\ 100 \\ \hline 320 \end{array}$$

100

χ^2

ranches, 25

Step 1:

Alternate hypothesis:

H₁:

	y_1	y_2	y_3	y_1^2	y_2^2	y_3^2
A	0	5	8	0	25	64
B	4	8	9	16	64	81
C	8	13	11	64	169	121
D	15	6	13	225	36	169
	$\Sigma y_1 = 27$	$\Sigma y_2 = 32$	$\Sigma y_3 = 51$	$\Sigma y_1^2 = 305$	$\Sigma y_2^2 = 294$	$\Sigma y_3^2 = 435$

Step 3: $N = 12$

Step 4: $T = \Sigma y_1 + \Sigma y_2 + \Sigma y_3$

$$= 27 + 32 + 51 = 110$$

Step 5: Correction factor $= \frac{T^2}{N}$

$$= \frac{110^2}{12} = 1008.3$$

Step 6:

$$T.S.S = \Sigma y_1^2 + \Sigma y_2^2 + \Sigma y_3^2 - C.F$$

$$= 305 + 294 + 435 - 1008.3$$

$$= 305 - 7$$

first
one

it

is

at

$$\begin{array}{r} 110 \\ 110 \\ \hline 000 \\ 100 \\ 100 \\ \hline 12100 \\ \hline 12100 \\ \hline 0 \end{array}$$

Step 7:

$$SSR = \frac{(\sum y_1)^2}{N_1} + \frac{(\sum y_2)^2}{N_1} + \frac{(\sum y_3)^2}{N_1}$$

$$= \frac{27^2}{4} + \frac{32^2}{4} + \frac{51^2}{4} = 1008.5$$

Step 8:

$$SSE = TSS - SSR$$

$$= 305.7 - 80.2$$

Step 9: ANOVA Table

Source of Variation	Sum of Squares	D.f	Mean Sum of Squares	Variance Ratio	F
Between rows	$SSR = 80.2$	$r-1 = 3-1 = 2$	$MSR = \frac{SSR}{r-1} = \frac{80.2}{2} = 40.1$	$F = \frac{MSR}{MSE}$	$F(2, 9) = 2.26$
Error	$SSE = 225.5$	$N-r = 18-3 = 9$	$MSE = \frac{SSE}{N-r} = \frac{225.5}{9} = 25.06$	$= \frac{40.1}{25.06} = 1.6$ (calculated value)	

Accept the

Step 10:

Here

and calculate

ie,

table value

hypothesis

He

difference

$$\frac{(\sum y_2)^2}{n_1} + \frac{(\sum y_3)^2}{n_1}$$

$$- C \cdot F$$

$$\frac{1}{1}^2 - 1008.3$$

$$0.2$$

Accept the null hypothesis.

Step 10:

Here the table value is 4.26

and calculated value = 1.6

i.e., calculated value is less than table value so we accept null hypothesis.

Hence there is no significant difference in brads.

Mean Sum of Squares	Variance Ratio	TV
MSR = $\frac{SSR}{\gamma - 1}$ = $\frac{80.2}{2}$ = 40.1	$F = \frac{MSR}{MSE}$ = $\frac{40.1}{2.506}$ = 1.6 (calculated value)	$F(2,9)$ = 4.26
MSE = $\frac{SSE}{9}$ = $\frac{22.5}{9}$ = 2.506		

② The following table shows that light in hours of 4 brands of electric lamps.

Brand	A	1610	1610	1650	1680	1700	1800	-
	B	1580	1640	1640	1700	1780	-	-
	C	1460	550	1600	1620	1640	1660	1740 1800
	D	1510	1520	1530	1570	1680	1680	-

Perform an analysis of variance the homogeneity of mean light of the 4 brands of the lamp.

Step 1: Null hypothesis:

There is no significant between brands.

Step 2: Alternative hypothesis:

There is ~~no~~ significant between brands.

Subtract 1
get X_1
 X_2
B
X₁ X₂

1	-2
1	4
5	4
8	10
10	15
12	1
2.0	-
ΣX_1	
= 57	

ble show
of 4 bran

1680	1700	1720
1700	1750	-
1620	1640	1660
1570	1680	1640

1800	-
-	-
1740	1820
-	-

ance the
mean
the

Subtract 1800 and then $\div$ by 10 we
get X_1, X_2, X_3, X_4 Total $X_1^2, X_2^2, X_3^2, X_4^2$

A	B	C	D					
X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
1	-2	-14	-9	-24	1	4	196	81
1	4	-5	-8	-8	1	16	25	64
5	4	0	-7	2	25	16	0	49
8	10	2	-3	17	64	100	4	9
10	15	4	0	29	100	225	16	0
12	-	6	8	26	144	-	36	64
12	-	14	-	34	144	-	196	-
20	-	22	-	42	400	-	484	-
-	-	-	-	-	-	-	-	-
ΣX_1	ΣX_2	ΣX_3	ΣX_4	$T =$	ΣX_1^2	ΣX_2^2	ΣX_3^2	ΣX_4^2
$= 57$	$= 31$	$= 29$	$= -19$	98	$= 735$	$= 361$	$= 957$	$= 267$

icant

is:

ficant

Step 3:

$$N = 26$$

Step 4:

$$T = 98$$

Step 5: Connection factor = $\frac{T^2}{N}$

$$= \frac{98^2}{26} \\ = 369.39$$

$$\text{Step 6: TSS} = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N} \\ = 1950.62 //$$

Step 7:

$$\text{SSR} = \frac{\sum X_1^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N} \\ = \frac{51^2}{7} + \frac{31^2}{5} + \frac{29^2}{8} + \frac{(19)^2}{6} - \frac{369.39}{1}$$

$$= 464.12$$

$$60.1 -$$

Step 8:

Step 9:

Source of Variation

Between Rows

Error

d

$$= 464 \cdot 12 + 192 \cdot 2 + 105 \cdot 12 + 60 \cdot 1 = 369.39$$

$$= 452.1711$$

Step 8: $SSB = TSS - SSR$

$$= 1950.62 - 452.1711$$

$$= 1498.349 //$$

Step 9: ANOVA TABLE.

Source of Variation	Sum of Squares	d.f	Mean Square	Variation Ratio
Between Rows	$SSC = 452.08$	$2-1 = 1$	$MSC = 452.08$	$F_{1,22} = 3.0$
Error	$SSE = 1498.36$	$N-2 = 22$	$MSE = 68.11$	

$$CV < TV$$

~~T~~ Tabulated ~~value~~ calculation value

So we accept ~~H0~~
 Hence there is no ~~diff~~ significant difference between ~~any~~ brand

Two way classification OR Randomised Block procedure

~~Design~~

① In order to compare 3 burners A, B, C 2 observation is made on each burner on 3 successive days. The data tabulated below

	A	B	C
Day 1	6	28	14
Day 2	28	36	38
Day 3	8	32	20

Perform an analysis of variance on these data & find whether difference between (i) the days (ii) the burner is significant at 5% level.

Step 1: n

between

Step 2:

days & b

Y_1

Y_2

Y_3

Σ

$=$

Step 3:

Step 4

Step 5:

OR Randomised

Step 1: null hypothesis:

There is no significant difference between days & burners.

Step 2: Alternative hypothesis:

There is significant difference between days & burners.

below

	X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
Y_1	6	28	14	48	36	784	196
Y_2	28	36	38	102	784	1296	1444
Y_3	8	32	20	60	64	1024	400
ΣX_1	ΣX_2	ΣX_3	ΣT	ΣX_1^2	ΣX_2^2	ΣX_3^2	
= 42	= 96	= 72	= 210	= 884	= 3104	= 2840	

Step 3:

$$N = 9$$

Step 4:

$$T = 210$$

Step 5:

$$\text{Correction factor} = \frac{T^2}{N} = \frac{210^2}{9} = 4900$$

Step 6:

$$\text{Correction factor} = \frac{T^2}{N}$$

Step 6:

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2$$
$$= CF$$

$$= 1128$$

Step 7:

$$SSR = \frac{\sum y_1^2}{3 N_1} + \frac{\sum y_2^2}{3 N_1} + \frac{\sum y_3^2}{3 N_1}$$
$$= \frac{T^2}{N}$$

$$= \frac{2304}{3} + \frac{10404}{3} + \frac{3600}{3}$$

$$= \frac{44700}{3}$$
$$= 14900$$

$$= 768 + 3456 + 1800$$

$$= 536$$

Step 8:

$$SSC =$$

$$= \frac{4}{3}$$

Step 9:

Step 10:

Source of variation

Between

columns

Between

Rows

Error

$$r = \frac{1}{N}$$

$$2 + \sum x_3^2$$

CF

Step 8:

$$SSC = \frac{(\sum x_1)^2}{N} + \frac{(\sum x_2)^2}{N} + \frac{(\sum x_3)^2}{N}$$

$$= \frac{42^2}{3} + \frac{96^2}{3} + \frac{72^2}{3} - \frac{T^2}{N}$$

$$= 488$$

$$+ \frac{\sum y_3^2}{3}$$

3N₁

Step 9:

$$SSE = TSS - SSR - SSC$$

$$= 1128 - 536 - 488$$

$$= 104$$

Step 10: ANOVA TABLE

Source of Variation	Sum of Squares	Design of freedom	Mean of Squares	Variance Ratio	Tv
Between Columns	SSC = 488	C-1 = 2	MSC = $\frac{SSC}{C-1}$	FC (b) = $\frac{MSC}{MSE}$	F(2,4) = 6.98
Between Rows	SSR = 536	R-1 = 2	MSR = $\frac{SSR}{R-1}$	FR = $\frac{MSR}{MSE}$	F(2,4) = 9.38
Error	SSE = 104	(C-1)(R-1) = 4	MSE = $\frac{SSE}{(C-1)(R-1)}$		

Step 11:

Here calculated value > table value

(i) We reject Null H_0 . There is a significant difference between burners

(ii) Here $L.V > T.V$ so we reject Null H_0 There is a significant difference.

days

H.W:

Q. Three tested in with 4 pounds a

A 6
C 8
B 1

Analyse state cor

Soln:

Variety

A
B
C

Total

Step 1:

between
slow m

2d value $\rightarrow$
table value

1. There is
no difference

2. Reject
null hypothesis

HW:

Q. Three varieties A, B, C of a crop are tested in a randomized block design with 4 replications. The plot yield in pounds are as follows.

A	6	C	5	A	8	B	9
C	8	A	4	B	6	C	9
B	7	B	6	C	10	A	6

Analyse the experimental yield and state conclusion.

Soln:

Variety	Block				Total
	X ₁	X ₂	X ₃	X ₄	
A	6	4	8	6	24
B	7	6	6	9	28
C	8	5	10	9	32
Total	21	15	24	24	84

Step 1:

There is no significant difference between column means as well as row means.

Step 2: There is significant difference between column means or the row means.

Step 3:

~~Step 3:~~

Step 4:

Variety	X ₁	X ₂	X ₃	X ₄	Total	Total ²
A	6	4	8	6	24	576
B	7	6	6	9	28	784
C	8	5	10	9	32	1024
Total	21	15	24	24	84	

Column $\sum X^2 = 624$
 $\sum 15^2 = 225$

Column Sequence:

$$21^2 = 441$$

$$15^2 = 225$$

$$24^2 = 576$$

$$24^2 = 576$$

$$\sum X^2 = 624$$

Step 3:

$$N = 3 \times 4$$

$$N = 12$$

Step 4:

$$T = 84$$

Step 5:

$$CF = \frac{T^2}{N}$$

Step 6: T

TSS

Step 7:

SSI

Step 8

SS

variance between
series.

Step 3:

$$N = 3 \times 4$$

$$N = 12$$

Step 4:

$$T = 84$$

Step 5: Correction factor

$$CF = \frac{T^2}{N} = \frac{84^2}{12} = \frac{7056}{12} = 588$$

Step 6: Total sum of squares

$$TSS = \sum X^2 - CF = 624 - 588 = 36$$

Step 7:

$$SSR = \frac{24^2}{4} + \frac{28^2}{4} + \frac{32^2}{4} - CF$$
$$= \frac{2384}{4} - 588 = 8$$

Step 8:

$$SSC = \frac{21^2 + 15^2 + 24^2 + 24^2}{3} - 588$$
$$= \frac{1818}{3} - 588 = 606 - 588 = 18$$

Step 9:

$$SSE = TSS - SSR - SSC$$

$$= 36 - 8 - 18 = 10$$

Step 10: ANOVA TABLE

Source of variation	Sum of Squares	Degree of freedom	Mean Square	F-ratio	TV null
Between varieties (Rows)	SSR 8	$c-1$ $3-1$ $= 2$	MSR SSR $= \frac{SSR}{r-1}$ $= \frac{08}{2} = 4$	FR (a, b) $\frac{MSR}{MSE}$ $= \frac{4}{1.667}$	FR $(2, 6)$ $= 5.14$
Between varieties (columns)	SSC 18	$r-1$ $= 4-1 = 3$	MSC $= \frac{SSC}{c-1}$ $= \frac{18}{3} = 6$	FC $= \frac{MSC}{MSE}$ $= \frac{6}{1.667} = 3.60$	$F_2(3, 6)$ $= 4.76$
Error	SSE 10	$(c-1)(r-1)$ $= 6$	mse $= \frac{SSE}{(r-1)(c-1)}$ $= \frac{10}{6} = 1.667$		

Step 11:

Here

(i) we take significant at

(ii) Here

Step 11:

Here Calculated value > Table value.

(i) we reject null H_0 . There is a significant difference between varieties

(ii) Here C.V. > T.V. so we reject

null H_0 . There is a significant difference between blocks.

n Fraction

12

FR

(2, 6)

$\frac{MSE}{MSE}$

$$= \frac{1}{1.667}$$

FR

(2, 6)

$$= 5.14$$

FC

$F_{2(3, 6)}$

$$= \frac{MSC}{MSE}$$

$$= 4.76$$

$$= \frac{6}{1.667}$$

$$= 3.60$$

Lating square (or) 3 way classification



①. Analyse the variance in the following Lating square yields of paddy (in kgs) where A, B, C, D denote the different cultivation method.

B20	C17	D25	A34
A23	D21	C15	B24
D24	A26	B21	C19
C26	B23	A27	D22

examine whether the different methods of cultivation have given significantly different in yields.

Step 1: H_0
Then

rows, columns

Steps: H_1
Then

between

treatment

Steps:

	X_i
Y_1	20.25
Y_2	-2
Y_3	-1
Y_4	1

ΣX

Step 3

Step 4

Classification:



in the
yields
D denote
othod.

Step 1: H_0

There is no significant difference between rows, columns & treatments.

Step 2: H_1

There is significant difference between rows / columns and treatments.

Step 3: Subtract 25

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	$20 - 25 = -5$	-8	0	9	$\sum Y_1 = -4$	25	64	0	81
Y_2	-2	-4	-10	-1	$\sum Y_2 = -17$	4	16	100	1
Y_3	-1	1	-4	-6	$\sum Y_3 = -10$	1	1	16	36
Y_4	1	-2	2	-3	$\sum Y_4 = -2$	1	4	4	9
	$\sum X_1 = -7$	$\sum X_2 = -13$	$\sum X_3 = -12$	$\sum X_4 = -1$	$\sum Y = -33$				
						31	85	120	127

Step 3:

$$N = 16$$

Step 4:

$$\bar{y} = -3.3$$

Step 5:

$$\text{Correction factor} = \frac{(-33)^2}{16} = \frac{1089}{16}$$

$$= 68.0625$$

$$= \frac{509}{4}$$

$$= \frac{509 - 1089}{4}$$

$$= 34.1875 //$$

Step 6:

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2$$

$$+ \sum X_4^2 - CF$$

$$= 31 + 85 + 120 + 127 - 68.0625$$

$$= 294.9375 //$$

Step 8:

$$85 = \frac{\sum X}{n}$$

$$= \frac{249}{4} + \frac{169}{4}$$

$$= 22.68$$

Step 7:

$$SSR = \frac{(\sum Y_1)^2}{N_1} + \frac{(\sum Y_2)^2}{N_1} + \frac{(\sum Y_3)^2}{N_1}$$

$$+ \frac{(\sum Y_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{16}{4} + \frac{289}{4} + \frac{100}{4} + \frac{4}{4} - 68.0625$$

Step 9:

SSE

$$= \frac{509}{4} - 68 \cdot 0625$$

$$= \frac{509}{4} - 68 \cdot 0625$$

$$= 34.1875 //$$

Steps:

$$SS = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_2} + \frac{(\sum X_3)^2}{N_3} + \frac{(\sum X_4)^2}{N_4} - \frac{T^2}{N}$$

$$= \frac{29}{4} + \frac{169}{4} + \frac{144}{4} + \frac{1}{4} - 68 \cdot 0625$$

$$= 22.6875$$

Step 4:

$$SSE = TSS - SSR - SSC$$

$$= 294.9375 - 34.1875 - 22.6875$$

Step 9;

					Total
A	9	-2	1	2	10
B	-5	-1	-4	-2	-12
C	-8	-10	-6	1	-23
D	0	-4	-1	-3	-8

$$SSK = \frac{10^2}{4} + \frac{(-12)^2}{4} + \frac{(-23)^2}{4} + \frac{(-8)^2}{4}$$

$$= 141.875$$

$$= 68.0625$$

Step 10:

$$SSE = TSS - SSR - SSC - SSK$$

$$= 96.875$$

Step 11.

Source of Variation

Between Columns

Between Rows

Between Treatments

Error

Step 1

The

best

(ii)

Ho

dis

(iii)

Then

tre

Step 11: ANOVA TABLE

Source of Variation	Sum of Squares	Df	Mean Squares	Variance Ratio	T.V
Between Columns	SSC = 28.675	(K-1) = 3	MSC = $\frac{SSC}{K-1}$ = 7.56	$F_C = \frac{16.145}{7.56}$ = 2.1357	$F_C (3, 9)$ = 4.76
Between Rows	SSR = 34.1875	(R-1) = 3	MSR = $\frac{SSR}{R-1}$ = 11.395	$F_R = \frac{16.145}{11.395}$ = 1.416	$F_R (3, 9)$ = 4.76
Between Treatment	SSK = 141.1875	(k-1) = 3	MSK = $\frac{SSK}{k-1}$ = 47.06	$F_K = \frac{16.145}{47.06}$ = 0.343	$F_K (3, 9)$ = 4.76
Error	SSE = 96.875	(R-1)(K-1) = 6	MSE = $\frac{SSE}{(R-1)(K-1)}$ = 16.145		

Step 12:

Conclusion:

i) $cal \cdot v < T \cdot v$

So we accept H_0

There is no significant difference between columns

(ii) $cal \cdot v < T \cdot v$ / so we accept

H_0 There is no significant difference between rows.

(iii) $cal \cdot v < T \cdot v$ / so we accept H_0
There is no significant difference between treatment

2) Examine whether the difference methods of ^{cultivation} have given significantly different yields.

S ₁₂₃	P ₁₂₃	R ₁₂₃	Q ₁₂₃
Q ₁₂₄	R ₁₂₃	R ₁₂₂	S ₁₂₅
P ₁₂₀	Q ₁₁₉	S ₁₂₀	R ₁₂₁
R ₁₂₂	S ₁₂₃	Q ₁₂₁	P ₁₂₂

Analyze the variance in the following square of yields (in kgs) paddy where P, Q, R, S denote the different methods of cultivation.

Soln:

Subtract 120 we get

S ₂	P ₁	R ₃	Q ₂
Q ₄	R ₃	P ₂	S ₅
P ₀	Q ₋₁	S ₀	R ₁
R ₂	S ₃	Q ₁	P ₂

	X ₁
Y ₁	2
Y ₂	4
Y ₃	0
Y ₄	8
Total	

Step 1: H₀
difference

treatment

Step 2: difference

Step 3

Step 4

Step 5

Step 6

2.21

difference

as

2

5)

3a

tion.

	X_1	X_2	X_3	X_4	Total
Y_1	2	1	3	2	8
Y_2	4	3	2	5	14
Y_3	0	-1	0	1	0
Y_4	2	3	1	2	8
Total	8	6	6	10	$T = 30$

Step 1: H_0 : there is no significant difference between columns and treatment.

Step 2: H_1 : There is significant difference between rows, columns and treatment.

Step 3: $N = 16$

Step 4: $T = 30$

Step 5: correction factor $= \frac{T^2}{N}$
 $= \frac{30^2}{16} = 56.25$

Step 6: $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2$
 $= 24 + 20 + 14 + 34 - 56.25$
 $= 35.75$

Step 7:

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{8^2}{4} + \frac{6^2}{4} + \frac{6^2}{4} + \frac{10^2}{4} - 56.25$$

$$= 2.75$$

Step 8:

(I have tran)

$$SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{8^2}{4} + \frac{6^2}{4} + \frac{6^2}{4} + \frac{10^2}{4} - 56.25$$

$$= 2.75$$

Step 9:

$$SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N} = 24.75$$

Step 9:

Ans

order a

P	0
Q	4
R	2
S	2

SSK =

Step 10

Step 10

$$+ \frac{(\sum x_3)^2}{N_1}$$

Step 9: SSK

Arrange the elements in the order of treatment

P	0	1	2	2	5
Q	4	-1	1	2	6
R	2	3	3	1	9
S	2	3	0	5	10

$$\frac{(\sum x_3)^2}{N_1}$$

$$SSK = \frac{5^2}{4} + \frac{6^2}{4} + \frac{9^2}{4} + \frac{10^2}{4} - 56.25$$

$$= 4.25$$

Step 10:

$$SSE = TSS - SSC - SSR - SSK$$

$$= 35.75 - 8.75 - 24.75 - 4.25$$

$$= 4$$

Step 11: ANOVA TABLE

Source of variation	Sum of square	d.f	Mean Square	Variance Ratio	T.V
Between columns	$SSC = 8.75$	$k-1 = 3$	$MSC = \frac{SSC}{k-1} = 0.92$	$F_C = \frac{MSC}{MSE} = 1.31$	$F_{(3,6)} = 4.76$
Between rows	$SSR = 24.75$	$k-1 = 3$	$MSR = \frac{SSR}{k-1} = 8.25$	$F_R = \frac{MSR}{MSE} = 12.31$	$F_R = (3,6) = 4.76$
Between treatment	$SSK = 4.25$	$k-1 = 3$	$MSK = \frac{SSK}{k-1} = 1.42$	$F_T = \frac{MSK}{MSE} = 2.19$	$F_T (3,6) = 4.76$
Error	$SSE = 4$	$(k-1)(k-1) = 6$	$MSE = \frac{SSE}{(k-1)(k-1)} = 0.67$	$F = \frac{MSC}{MSE} = 1.31$	

Step II: Conclusion.

(i) $C.V.F_C < T.V.F_C$ we accept H_0
There is no significant difference in columns.

(ii) $C.V.F_R < T.V.F_R$ we accept H_0
There is no significant difference in Rows.

There
differs

n Variance
Ratio

TV

$$F_2 = \frac{MSE}{MSE} = 1.31$$

$$F_{(3|6)} = 4.76$$

$$F_2 = \frac{MSE}{MSE} = 12.31$$

$$F_{(3|6)} = 4.76$$

$$F_2 = 4.2$$

$$F_{(3|6)} = 4.76$$

$$F_2 = \frac{MSE}{MSE}$$

$$F_2 = 2.119$$

(iii) $C.V.F_t < T.V.F_t$ we accept H_0 .
There is no significant
difference between treatment.

exp 110

error in

accept

diff

Unit - 4:

Sample:

A finite subset of statistical individuals in a population is called Sample.

Sample Size:

The no of individuals in a sample is called Sample Size.

Sampling Error:

On examine a sample of a particular stuff be arrived at a decision of purchasing or rejecting stuff the error involved in such approximation is known as Sampling error.

statistical
on is

note...

if we want to test the
null hypothesis that the population
has a specified mean μ_0 . (say)
 $H_0: \mu = \mu_0$. Then the alternate

in a
test.

hypothesis.

i) $H_1: \mu \neq \mu_0$ (Two-tailed)

ii) $H_1: \mu > \mu_0$ (Right tailed)

iii) $H_1: \mu < \mu_0$ (Left tailed)

Two types of error:

i) Type I error: Reject H_0 .
when it is true.

(ii) Type II error: Accept H_0
when it is wrong

Small Samples ($n < 30$).

When the size of the

sample $n < 30$ then the sample is
called Small Sample.

stuff
of

of the
maximum
error.

Student's t - test ::

$$\text{Test statistic } t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

(or)

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

Confidence or fiducial limit for μ :

if $t_{0.05}$ is the table value of t for $n-1$ degrees of freedom at 5% level of significance then 95% confidence limit

$$\text{for } \mu = \bar{x} \pm t_{0.05} \frac{s}{\sqrt{n}}$$

99% confidence limit for $\mu = \bar{x} \pm t_{0.01} \frac{s}{\sqrt{n}}$

The mean of life time of a sample of 25 fluorescent light bulbs produced by a company is computed to be 1570 hours with the S.D of

120 hours. The company claims the avg life of bulbs produced

by the company is 1600 hours

using the level of significance of 0.05 is a claim acceptable?

$\mu \rightarrow$ population mean

$\bar{X} \rightarrow$ sample mean

soln:

$$\bar{X} = 1570$$

$$\mu = 1600$$

$$n = 25 < 30$$

$$S.D = 120$$

Degree of freedom
 $= n - 1$
 $= 25 - 1$
 $= 24$

$$D = 1570 - 1600 = -30$$

$\frac{-30}{120/\sqrt{25-1}} = -2.5$

Null hypothesis H_0 : The claim is acceptable.

Alternate Hypothesis.

H_1 : the Claim is not acceptable

i.e) $\mu \neq 1600$.

Wkt:

$$\text{Test Statistic } t = \frac{\bar{x} - \mu}{S.D / \sqrt{n-1}}$$

$$= \frac{1570 - 1600}{120 / \sqrt{24}}$$

$$= -30$$

$$120 / \sqrt{24}$$

$$= -1.22$$

$$t = -1.22$$

$$|t| = |-1.22|$$

$$t = 1.22$$

$\therefore$ the C.V of $t = 1.22$

The table
significant
for

C.V of

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Hence

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range

mean:

10 boy

The table value of t at 5% level of significance with 24 ^{degree} of freedom for 2 tailed test $\Rightarrow 2.064$

$$C.V \text{ of } t < T.V \text{ of } t$$

so we accept null hypothesis.

Hence the claim is acceptable.

②. The random sample of 10 boys had the following IQ is 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. do these data support the assumption of a population mean IQ of 100. find the reasonable range in which most of the mean IQ values of sample like 10 boys.

Soln:

here S.D & mean of the sample is not given directly.

$$S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$S^2 = \frac{1}{9} \sum (x_i - \bar{x})^2$$

$$\bar{x} = 97.2 \text{ (avg)}$$

x	$(x - \bar{x})$	$(x - \bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	9.2	84.64
83	14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84

~~S^2~~

Given

mean

Alt

Te

$$s^2 = \frac{1}{9} (1833.60) = 203.73$$

Given that

$$n = 10$$

$$\mu = 100$$

~~H0 hypothesis~~

null H0. The data support the assumption of population mean μ of 100.

Alternative H0 $H_1: \mu \neq 100$.

$$\therefore s^2 = \frac{1}{9} (1833.60)$$

$$= 203.73$$

$$s = 14.273$$

$$\text{Test Statistic } z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{97.2 - 100}{14.27/\sqrt{10}}$$

$$z = -0.62$$

$$|t| = 0.62$$

The calculated value of t
 $= 0.62$

The table value of t at
 5% level of significance,
 $= 2.262$ (Table)

We accept the C.V of $t < T.V$
 of t therefore we accept null
 hypothesis.

The data supports the assumption
 of mean IQ 100 in the population
 that is 95% confidence limit

$$= \bar{x} \pm t_{0.05} \times \frac{s}{\sqrt{n}}$$

$$= 97.2 \pm 2.262 \times \frac{14.27}{\sqrt{10}}$$

$$= 97$$

$$= 100$$

when
 of 10

Student
mean

$$s^2$$

f &

$$= 97.2 \pm 10.207$$

$$= 107.407 \text{ (or) } 86.93$$

at

$\therefore$ the 95% confidence limit where the mean values sample of 10 boys will lie in (86.93,

$\frac{1}{2} T.V$

u

Student's t-test for difference of

means



$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{1/n_1 + 1/n_2}}$$

upt to

extra

$$S^2 = \frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

limit

$$\text{(or) } n_1 + n_2 - 2$$

$$S^2 = \frac{1}{n_1 + n_2 - 2} [n_1 s_1^2 + n_2 s_2^2]$$

Q. Two horses A and B were tested according to the type (in seconds) to run a particular track with a following results

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	27	-

(no difference)

Test whether you can discriminate between two horses. you can use the fact that 5% value of t for 11 degrees of freedom is 2.22.

Solution

NULL Hypothesis H_0 : There is no significant difference between two horses

Alternative Hypothesis H_1 : $\mu_1 \neq \mu_2$

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12

x_1	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	x_2	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
28	-3.28	10.7584	29	1.17	1.3689
30	-1.28	1.6384	30	2.17	4.7089
32	0.72	0.5184	32	2.17	4.7089
33	1.72	2.9584	34	-3.83	14.6689
33	1.72	2.9584	37	0.83	0.6889
29	-2.28	5.1984	27	0.83	0.6889
34	2.72	7.3984	-	0.83	0.6889
299	0.04	31.4288	167	0.02	26.8336

$$s^2 = 31.4288 + 26.8336$$

$$\frac{31.4288 + 26.8336 - 2}{7 + 6}$$

$$= \frac{31.4288 + 26.8336}{11 - 2}$$

$$= 5.2965$$

$$s = 2.3015$$

$$t = \frac{31.28 - 27.80}{\sqrt{1/7 + 1/6}} = 2.86$$

$$\text{Answer } 2.3015 \sqrt{1/7 + 1/6}$$

F-test

To test, if the two surface have come from the same population we use F-test.

Condition:

Null Hypothesis $H_0: \sigma_1^2 = \sigma_2^2$ (The population are same). The test

statistics $F = \frac{S_1^2}{S_2^2}$ - $S_1 \rightarrow \text{big} \rightarrow \text{Numerator}$
 $S_2 \rightarrow \text{small}$.

where

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$$

$$S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

Degrees of freedom

$$V_1 = n_1 - 1$$

$$V_2 = n_2 - 1$$

①. The two surface be as

Can it come from population

So test from it

(i) F-test
(ii)

stu

①. The nicotine content in milligrams in two samples of tobacco were found to be as follows sample A, sample B

Sample A	24	27	26	21	25
Sample B	27	30	28	31	23

Can it be said that 2 samples come from the same normal population.

mean - t test
variance - F test

$$\therefore n_1 = 5, n_2 = 6$$

So test whether the 2 samples come from the same normal population.

(i) equality of variance by using F-test.

(ii) equality of mean by using student t-test.

x	$x - \bar{x}$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$
24	-0.6	0.36	27	-2	4
27	2.4	5.76	30	1	1
26	1.4	1.96	28	-1	1
21	-3.6	12.96	31	2	4
25	0.4	0.16	22	-7	49
<u>123</u>		<u>21.2</u>	<u>36</u>	<u>7</u>	<u>49</u>
		174			108

The
Degree

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{21.2}{4} = 5.3$$

$$S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{108}{5} = 21.6$$

The
Deg
of

So

That

(i) F test

Null Hypothesis $H_0: \sigma_1^2 = \sigma_2^2$

(ii)

Test Statistic $F = \frac{S_2^2}{S_1^2}$ (big)

$$= \frac{21.6}{5.3}$$

$$= 4.07$$

the C.V of $F = 4.07$
Degree of freedom:

$$= (n_2 - 1, n_1 - 1)$$

$$= (5, 4)$$

$$= 6.26 \text{ (T.V)}$$

the T.V. of F for (5, 4)
Degree of freedom at 5% level
of significance is 6.26.

$\therefore$ C.V of $F <$ T.V F

So we accept the null H_0 .

That is the variance are equal.

$$(ii) \quad S = \frac{1}{9} [(5)(5.3) + (6)(11.6)]$$

$$= \frac{1}{9} \times 115.6$$

$$= 12.84$$

$$S = 4.15$$

$$t = \frac{24.6 - 29}{4.15 \sqrt{\frac{1}{5} + \frac{1}{6}}}$$

$$= \frac{-4.4}{4.15 \times 0.36}$$

$$= \frac{-4.4}{1.494}$$

$$= -1.92$$

t test : $(n_1 + n_2 - 2)$
 $(9, 5) \rightarrow 0.05 \rightarrow 2.262$

②. 2 Random Sample give the following results

Sample	Size	Sample Mean	Sum of square of deviation from the mean
1	10	15	90
2	12	14	108

to test whether the two samples come from the same normal population.
soln:

The null H_0 : The two samples drawn from the same normal population.

$$n_1 = 10, n_2 = 10$$

To test whether the 2 samples come from the same normal population we have to test (i) equality of variance by F -test
(ii) equality of mean by t -test.

$$\sum (x_1 - \bar{x}_1)^2 = 90$$

$$\sum (x_2 - \bar{x}_2)^2 = 108$$

find $S_1^2 = \frac{\sum (x_1 - \bar{x})^2}{n_1 - 1}$

$$= \frac{90}{9} = 10$$

$$S^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{108}{9}$$

$$= 12$$

(i) F-Test

$$\text{Null } H_0: \sigma_1^2 = \sigma_2^2$$

$$\text{Test Statistic } F = \frac{S_1^2}{S_2^2}$$

$$= \frac{10.1}{9.81}$$

$$= 1.019$$

The C.V. of 1.019

Degrees of freedom

$$(n_1 - 1, n_2 - 1)$$

$$(9, 11)$$

$$= 2.90$$

T.V of F for (9, 11) Degrees of freedom is 2.90

$$C.V < T.V$$

So we accept the Null H_0 .
This is the variance are equal

$$(ii) \bar{X}_1 = 10$$

$$\bar{X}_2 = 1$$

$$\sum (x_1 - \bar{X}_1)^2$$

$$\sum (x_2 - \bar{X}_2)^2$$

$$S^2 = \frac{1}{n_1}$$

$$= \frac{1}{20}$$

$$= \frac{1}{20}$$

$$= \frac{1}{20}$$

$$= \frac{1}{20}$$

$$= \frac{1}{20}$$

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$$= \frac{1}{20}$$

$$= \frac{1}{20}$$

$$s_2^2$$

$$= s_1^2$$

$$s_2^2$$

$$10$$

$$9.81$$

$$1.019$$

$$1.019$$

$$1.019$$

$$1.019$$

$$1.019$$

$$1.019$$

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$$1.019$$

$$1.019$$

$$\bar{x}_1 = 15$$

$$\bar{x}_2 = 14$$

$$\sum (x_1 - \bar{x}_1)^2 = 90$$

$$\sum (x_2 - \bar{x}_2)^2 = 108$$

$$s^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2 \right]$$

$$= \frac{1}{20} [90 + 108]$$

$$= \frac{1}{20} \times 198 = 9.9$$

$$s = 3.146$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{1/n_1 + 1/n_2}} = \frac{15 - 14}{3.14 \sqrt{1/10 + 1/12}}$$

$$= \frac{1}{3.14 \sqrt{1/10 + 1/12}} \rightarrow \frac{(15 - 14) \div (3.14 \sqrt{(1/10 + 1/12)})}{1} = 0.46$$

$$\begin{aligned}\text{Degrees of freedom} &= n_1 + n_2 - 2 \\ &= 10 + 12 - 2 \\ &= 20\end{aligned}$$

The T.V of t for 20 degrees of freedom at 5% level of significance

$$is \ 2.086$$

$$C.V < T.V$$

So we accept null H₀.

Hence the Sample comes from normal population.

Large Samples..

If the size of the sample n is greater than 30, then the sample is called large sample

* * T
* 90
between
the

w
Note:
propo

①. In
book n
Thus
come
men

$$n_1 + n_2 - 2$$

$$10 + 10 - 2$$

$$= 20$$

20 degrees of
freedom of significance

and H.O.

comes from

• The test of significance of single
sample test the significant difference
between the sample proportion and
the test statistic $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

$$\sqrt{\frac{PQ}{n}}$$

Where $Q = 1 - P$.

Note:

Limit for the population
proportion P is given by

$$p \pm 3 \sqrt{\frac{PQ}{n}}$$

① In the bigity 325 men out of
600 men were found to be smokers.
Thus this information supports the
conclusion that the majority of
men are smokers:

Soln:

$$n = 600$$

$P \rightarrow$ Sample
 $P \rightarrow$ population

$$\text{no of smokers} = 325$$

$P =$ Sample proportion in smokers

$$= \frac{325}{600} = 0.5417$$

$P =$ population proportion of smokers

$$= 1/2$$

$$Q = 1 - P = 1/2$$

Null H_0 : The no. of smokers & Non Smokers are equal in the cities.

Alternative H_1 : P is greater than $\frac{1}{2}$.

Test stat:

$$= \frac{0.5417 - \frac{1}{2}}{\sqrt{\frac{0.5 \times 0.5}{600}}}$$

The calculated level of 5% tailed t

Since t $t.v$ of z

the no. smokers

②. Ac and of

3 or 4

with the was us

$$\begin{aligned} \text{test statistic} &= z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \\ &= \frac{0.5417 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 1.6457 \end{aligned}$$

the calculated value of $z_{0.05}$ level of significance for the right tailed test. 1.645 .

Since the C.V of z greater than T.V of z so we reject the null H_0 .

the majority of men are in city smokers.

②. A die was thrown 9000 times.

and of these 3220 yield at a

3 or 4. Is this consistent

with the H₀ that the die was unbiased

Given

$$n = 9000$$

p = proportion of success of getting 3 or 4 in 9000 throws

$$= \frac{3220}{9000} = 0.357$$

P = Proportion of success in the population

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} \approx 0.333$$

$$Q = 1 - P = 0.677$$

Null H_0 : The die is unbiased

Alternative H_1 : The die is ^{biased} ~~biased~~ (2 tailed)

$$Z = \frac{0.357 - 0.333}{\sqrt{\frac{0.333 \times 0.677}{9000}}}$$

$$= \frac{0.024}{\sqrt{\frac{0.333 \times 0.677}{9000}}}$$

$$= 4.94$$

The ca

5%

tailed

c.v of

80

∴

Differ

differen

P_1, P_2

where

of getting
three.

The calculated value of z at 5% level of significant for two tailed test.

$$z = 3$$

n the

C.V of $z \geq T.V$ of z .

So we reject the H_0 .

$\therefore$ the die is biased.

Difference of Proportions:

To test the significant difference between the sample proportions

P_1, P_2 then test statistics $z = \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

where $P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$.

$$Q = 1 - P.$$

①. Random Sample of 400 men and 600 women were asked to whether they would like to have a ~~friend~~ ^{partner} near their residence. 200 men & 325 women were in favour of the proposal. Test the H_0 hypothesis that proportion men & women in favour are same at 5% level.

Soln:

given, $n_1 = 400$

$n_2 = 600$

$P_1 = \text{proportion of men} = \frac{200}{400} = 0.5$

$P_2 = \text{proportion of women}$

$$= \frac{325}{600} = 0.541$$

H_0 : The proportion of men & women in favour of proposal are same. $(P_1 = P_2)$

Alternative H_0 : The proportion of women in favour of P_1 are not same ($P_1 \neq P_2$)

Test statistic

$$Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$= \frac{400 \times 0.5 + 600 \times 0.5}{400 + 600}$$

$$= 0.525$$

$$Q = 1 - P = 1 - 0.525 = 0.475$$

$$Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\sqrt{0.525(0.475) \left(\frac{1}{400} + \frac{1}{600} \right)}$$

$$= -1.645$$

$$= 0.041$$

$$\sqrt{0.525(0.475) \left(\frac{1}{400} + \frac{1}{600} \right)}$$

$$\left(\frac{1}{400} + \frac{1}{600} \right)$$

C.V

2) In two large populations there are 30% and 25% respectively fair haired people. In this difference likely to be hidden in samples of 1200 and 900 respectively from the 2 population.

Given, $n_1 = 1200$, $n_2 = 900$

P_1 = proportion of first fair hair people $= \frac{30}{100} = 0.3$

P_2 = proportion of second fair hair people $= \frac{25}{100} = 0.25$

Null H₀

Altern

Test

$$= \frac{a}{n}$$

$$= \sqrt{\frac{a}{n}}$$

$$=$$

Two.

Null H_0 : Assume that the sample proportion are equal.

Alternative H_1 : The difference population proportion likely ^{population} sample.

Test statistics $Z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$ $P_1 \neq P_2$

$$\begin{aligned} Q_1 &= 1 - P_1, Q_2 = 1 - P_2 \\ Q_1 &= 1 - 0.3 = 0.7 \rightarrow Q_2 = 1 - 0.25 = 0.75 \\ &= 0.3 - 0.25 \end{aligned}$$

$$\sqrt{\frac{(0.3)(0.7)}{1200} + \frac{(0.25)(0.75)}{900}}$$

$$= 2.55$$

$$T.V = 1.96$$

Two tailed

We reject the null H_0 .

Test of significant for difference of mean:

Mean	Size	Population	Variance
$\bar{x}_1$	n_1	μ_1	σ_1^2
$\bar{x}_2$	n_2	μ_2	σ_2^2

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Mean: $\bar{x}_1, \bar{x}_2$

Size: n_1, n_2

population: μ_1, μ_2

Variance: σ_1^2, σ_2^2

Samples have drawn from same population $\sigma_1^2 = \sigma_2^2 = \sigma$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

of σ^2

① - The
6400 e
170cm

if σ is not known we can estimate
of σ^2 given by $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$

① - The simple sample n_1 of n_2 heights of
6400 english men have a mean of
170cm and SD of 8.4 cm

~~chi square~~

Chi square test of goodness of fit.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O $\rightarrow$ Observed frequency.

E $\rightarrow$ Expected frequency.

If the data is given the degrees of freedom = $n - 1$ for the binomial Distribution

Poisson distribution The degree of freedom = $n - 2$

for the normal distribution $= n-3$.

The theory predicts the proportion of
③ ~~4 groups~~ beads in the 4 groups

A, B, C, D 9:3:3:1. In an experiment
with 1600 beads. The numbers in the
four groups were 882, 313, 287
and 118. Does the experiment results
support the theory. (Given $\chi^2_{0.5} = 7.81$)

Soln:

for 3d.f]

NULL H₀: The experimental results support
the theory

Divide 1600 in the ratio 9:3:3:1

$$9 + 3 + 3 + 1 = 16$$

$$\frac{1600}{16} = 100$$

$$9 \times 100 = 900$$

$$3 \times 100 = 300$$

$$3 \times 100 = 300$$

$$1 \times 100 = 100$$

we get expected 900, 300, 300, 100

Observed frequency	Expected frequency (E)	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
(0)	900	-18	324	0.36
882	800	13	169	0.56
313	800	-13	169	0.56
287	100	+18	324	0.36
118				
				4.72

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= 4.72$$

The C.V of $\chi^2 = 4.726$ The T.V of χ^2 for 3 degrees of freedom at 5% level of significance is 7.81

C.V < T.V so we accepted the null

②. Then no ... the v ... test ...
Days |
no of accident |
NULH
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Obsor
freq
0
14
18
18

Q. The following table gives the no. of accident that occur during the various day of the week. Test whether the accidents are ~~uniformly~~ uniformly over the week.

Days	Mon	Tue	Wed	Thu	Fri	Sat
no. of accident	14	18	12	11	15	14

SOLUTION:
The accidents are uniformly distributed over the week.

The expected frequencies of accidents on each of the days are,

$$\frac{14+18+12+11+15+14}{6} = \frac{84}{6} = 14$$

Observed frequency	Expected frequency	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
0	14	0	0	0
14	14	4	16	1.14
18	14	-2	4	0.286
12	14	2	4	0.286

11	14	-3	9	0.66
15	14	1	1	0.07
14	14	0	0	0

$$C.V = 2.134$$

The C.V of $\chi^2 = 2.134$

The table value of $\chi^2 =$

The Degrees of freedom

$$n - 1 = 6 - 1 = 5$$

11.070 $\rightarrow$ T.V at 5% level of significance

$$C.V < T.V$$

So we accept the Null Ho.

$\downarrow$
 other full ab
 lution

Formula:
 $\chi^2 = \sum \frac{(O-E)^2}{E}$

of Attribute

A
B

Exp

H.W

The fo
 distribution
 Random

Digits

frequency

Null H

frequency

0.663
0.041
0

✓ 2.134

Formula
Chi-square Test for Independence of Attributes.

A	a	d
B	c	d

a	b	a+b
c	d	c+d
a+c	b+d	N

Expected frequencies:

$E(a) = \frac{(a+c)(a+b)}{N}$	$E(b) = \frac{(b+d)(a+b)}{N}$	a+b
$E(c) = \frac{(a+c)(c+d)}{N}$	$E(d) = \frac{(b+d)(c+d)}{N}$	c+d
a+c	b+d	N

H.W

The following figures show the distribution of digits in numbers chosen at random from a telephone directory.

Digits	0	1	2	3	4	5	6	7
frequency	1026	1107	997	966	1075	933	1107	972

8 9
964 850

NOTE: The digits occur equally frequent in the directory

Digits	Observed frequency (O)	Expected frequency (E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
0	1026	1000	676	0.676
1	1107	1000	11449	11.449
2	997	1000	9	0.009
3	966	1000	1156	1.156
4	1075	1000	5625	5.625
5	933	1000	4489	4.489
6	1107	1000	11449	11.449
7	972	1000	784	0.784
8	964	1000	1296	1.296
9	853	1000	21609	21.609
	10000	10,000		58.54

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 58.542$$

$$C.V = 58.542$$

$$E.V \text{ of } \chi^2 =$$

The degree of freedom

$$= n - 1$$

$$= 10 - 1 = 9$$

At 5% level

so

frequency

② The classification to see work

mat
Fres

so test work of to

Digits	Observed frequency (O)	Expected frequency (E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
0	1026	1000	676	0.676
1	1107	1000	11449	11.449
2	997	1000	9	0.009
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$$\chi^2 = \sum \frac{(O-E)^2}{E} = 58.542$$

$$C.V = 58.542$$

$$E.V \text{ of } \chi^2 =$$

The degree of freedom
 $= n - 1$
 $= 10 - 1 = 9$

At 5% Significance

so the

frequent

② The following classification to sections works

male
female

no test was work is in of the work

$$\chi^2 = \frac{(O-E)^2}{E}$$

676	0.676
11449	11.449
9	0.009
1156	1.156
5625	5.625
4489	4.489
11449	11.449
784	0.784
1296	1.296
21609	21.609
58.58	

At 5% significant level.

$$= 16.919$$

$$\therefore C.V > T.V$$

We reject the H_0

so the digit do not occur equally frequently in the telephone history.

② The following table gives the classification of 100 workers according to ~~section~~ sex and nature of works.

	Stable	unstable	total
male	40	20	60
Female	10	30	40
	50	50	100

To test whether the nature of work is independent of the sex of the workers.

NOTE: The nature of the work is independent of the sex of the workers.

The expected frequency given in the table:

$\frac{50 \times 60}{100} = 30$	$\frac{50 \times 60}{100} = 30$	60
$\frac{50 \times 40}{100} = 20$	$\frac{50 \times 40}{100} = 20$	40
50	50	100

Observed frequency (O)	Expected frequency E	O-E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
10	30	-20	400	13.33
40	30	10	100	3.33
20	20	0	0	0
10	20	-10	100	5
30	20	10	100	5
				<u>16.66</u>

$$\chi^2 = 2$$

then deg

at

We

③. In cattle resul

Included

not

involved

$$\chi^2 = \sum 16.667 \rightarrow C.V$$

$$\begin{aligned} \text{the degree of freedom} &= (r-1)(c-1) \\ &= (2-1)(2-1) \\ &= 1 \end{aligned}$$

at 5% level significance

$$3.841$$

$$C.V > T.V$$

We reject the H_0 .

③. In an experiment on immune cattle for tuberculosis the following results were obtained.

	affected	Not affected
Involved	12	26
not involved	10	6

NOTE: The effect of vaccine
in control of tuberculosis.

58×38 $60 = 11.7$	$32 \times 28 = 20.2$ 60	38
28×22 $60 = 10.2$	32×22 $60 = 11.7$	22
28	32	60

Observed frequency	Expected frequency	$(O-E)$	$(O-E)^2$	$(O-E)^2/E$
(0)	17.73	-5.73	32.83	1.85
12	20.26	5.74	32.94	1.625
26	10.26	5.74	32.94	3.20
16	11.73	-5.73	32.83	2.79
6				9.484

$$\chi^2 = 9.484$$

$$D.O.F = (n_1 - 1)(n_2 - 1) = 1$$

$$(2-1)(2-1)$$

a line

$$S \cdot t-v \text{ at } 5\% = 3.841$$

so we reject the null.

$$\chi^2 (O-E)^2$$

E

$$3 \quad 1.85$$

$$4 \quad 1.625$$

$$1 \quad 3.27$$

$$3 \quad 2.74$$

$$9.984$$

Bay's theorem:

$$P(B_i|A) = \frac{P(B_i) P(A|B_i)}{\sum_{j=1}^n P(B_j) P(A|B_j)}$$

①. Companies B_1, B_2, B_3 produce 30%, 45%, 25% cars respectively. It is known that 2%, 3% and 2% of these cars produced are defective.

(i) What is the probability that a car purchased is defective?

(ii) If a car purchased is found to be defective what is the probability that this car is produced by company?

$$P(B_1) = \frac{30}{100}, P(B_2) = \frac{45}{100}, P(B_3) = \frac{25}{100}$$

$A \rightarrow$ defective

$$P(A|B_1) = \frac{2}{100}; P(A|B_2) = \frac{3}{100}; P(A|B_3) = \frac{2}{100}$$

$$P(A) = P(B_1) P(A|B_1) + P(B_2) P(A|B_2) + P(B_3) P(A|B_3)$$

$$= \frac{30}{100} \times \frac{2}{100} + \frac{45}{100} \times \frac{3}{100} + \frac{25}{100} \times \frac{2}{100}$$

$$K^2 + A =$$

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